

Viscereality: Bioresponsive Virtual Reality for Box Breathing and Altered-State Experience

George Fejer^{1,2,3*}, Till Holzapfel⁴, Taru Hirvonen,
Anestis Lalidis Mateo⁶, Johannes Blum⁵, Michael Gaebler²,
Bigna Lenggenhager^{7,8}

^{1*}University of Konstanz, Konstanz, Germany.

²Max Planck Institute for Human Cognitive and Brain Sciences,
Leipzig, Germany.

³ALIUS Research Network, Berlin, Germany.

⁴University of Duisburg-Essen, Research Center Trustworthy Data
Science and Security, Duisburg, Germany.

⁵Pädagogische Hochschule Schaffhausen, Schaffhausen, Switzerland.

⁶Institute for Building Intelligence, Hochschule Bielefeld - University of
Applied Sciences and Arts, Bielefeld, Germany.

⁷AIR—Association for Independent Research, Zurich, Switzerland.

⁸Department of Psychology, University of Zurich, Zurich, Switzerland.

*Corresponding author(s). E-mail(s): george.fejer@aliusresearch.org;
Contributing authors: tillholzapfel@gmail.com; 2taruhi@gmail.com;
alalidis@hsbi.de; johannes.blum@phsh.ch; gaebler@mpib-berlin.mpg.de;
bigna.lenggenhager@gmail.com;

Abstract

Breath-based interactions in virtual reality typically map breathing onto discrete visual objects or physiological proxies. Viscereality instead translates respiratory state into a surrounding particle field, mapping lung volume onto external space so that inhalation expands the environment and exhalation contracts it. Because particle fields have no fixed boundary, they can convey information through the motion and interactions of many simple elements. We tested whether this virtual-reality paradigm could reliably elicit altered-state experiences during box breathing, whether transient symmetric particle coupling would further influence subjective experience, and whether participants could identify the manipulation.

In a preregistered within-subjects study, 39 participants completed three counterbalanced 10-minute audio-guided box-breathing sessions: a high-symmetry virtual-reality condition, an asymmetric virtual-reality condition, and a dark-screen control. We assessed breathing adherence, altered states of consciousness, pictographic self-related measures, and sensitivity to the symmetry manipulation. Relative to the dark-screen control, the pooled virtual-reality conditions improved breathing synchronization, increased ratings of the perceived spatial extension of the self, and produced altered-state profiles with strong Perceptual Effects, moderate Positive Effects, and weaker Distressing Effects. Participants detected some symmetry-relevant features above chance, but neither reliably identified which session was symmetric nor showed outcome differences between the two virtual-reality mappings. These findings show that the overall virtual-reality paradigm produced reliable experiential differences from the control, but they do not establish that bioresponsiveness rather than visual engagement caused those differences. The specific affective contribution of particle symmetry, and whether it requires explicit awareness, therefore remain unresolved.

Keywords: biofeedback, breath-based interaction, breathwork, altered states of consciousness, particle systems

1 Introduction

1.1 Breath as an Affective Sensory Modality

Breathing carries both interoceptive and exteroceptive sensory information. Its interoceptive component provides information about the body’s internal respiratory state, including lung inflation, airway and respiratory-muscle activity, and oxygen and carbon dioxide levels (Chan et al. 2024; Engelen et al. 2023; Weng et al. 2021). Because inhaled air stems from the external environment, breathing also samples exteroceptive information about ambient conditions, including odours, airflow, and air temperature, through its coupling with orthonasal olfaction and thermo- and mechanosensory receptors in the upper airways (Mori and Sakano 2022a,b; Zhou et al. 2011; Sabnis et al. 2008; Kytikova et al. 2021). Breathing is also closely tied to affective experience. People sigh with relief after stress (Balban et al. 2023), hold their breath during freeze responses (Boiten et al. 1994), and hyperventilate during fear or panic (Jerath and Beveridge 2020; Kreibig 2010). Arousing negative emotions such as fear and anger are linked to shallower, more rapid breathing, whereas relaxation and contentment correspond to slower, deeper respiratory patterns (Boiten et al. 1994; Homma and Masaoka 2008; Jerath and Beveridge 2020). Unlike most visceral signals such as heartbeat, breathing is also amenable to voluntary control (Sarkar 2017), making this relationship bidirectional. Voluntarily changing respiratory patterns can modulate affective state (Jerath and Beveridge 2020; Ashhad et al. 2022), a principle that underlies breathwork as a therapeutic modality (Fincham et al. 2023; Balban et al. 2023; Siebieszuk

George Fejer and Till Holzapfel contributed equally and share first authorship. Taru Hirvonen and Anestis Lalidis Mateo contributed equally and share second authorship. Johannes Blum is the third author. Michael Gaebler and Bigna Lenggenhager share senior authorship.

et al. 2025). Breathwork techniques range from slow-paced protocols that target relaxation, stress regulation, and improvements in heart rate variability through sustained cardiovascular-respiratory coupling (Balban et al. 2023; Ahmed et al. 2021; Siebieszuk et al. 2025), to high-ventilation protocols such as cyclic hyperventilation that can facilitate more pronounced alterations in affect and consciousness (Bahi et al. 2024; Lewis-Healey et al. 2024; Havenith et al. 2025), effects increasingly posited as therapeutic, for instance in facilitating emotional breakthroughs (Fincham et al. 2026, 2024). Breathwork is already established as a therapeutic modality for stress, anxiety, and depression (Fincham et al. 2023; Bentley et al. 2023; Banushi et al. 2023), and a growing number of virtual-reality applications also incorporate breathing for therapeutic and health-related purposes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). Yet this literature has so far explored only a relatively limited portion of the broader design space that breathing affords as an interaction modality in VR.

1.2 Breath-Based Interactions in VR

Existing breath-based VR systems span meditation (Prpa et al. 2020; Shamekhi and Bickmore 2018), anxiety regulation (Bossenbroek et al. 2020), stress management through heart-rate-variability biofeedback (Blum et al. 2019; Rockstroh et al. 2019), and rehabilitation (Shi et al. 2023; Alhammad 2024). In the context of our framework, we distinguish between two motifs in the design space regarding the mapping of the user’s respiratory signal onto discrete visual objects, in contrast to environmental mappings. Object-oriented approaches typically place respiratory feedback in a discrete visual target. Third-person variants let the user watch an external entity respond to breathing, such as a cloud that approaches during inhalation and recedes during exhalation (Tinga et al. 2019). In first-person variants, the mapping is applied to a virtual body seen from the user’s own vantage point. The “embreathment” illusion demonstrated that synchronising a virtual avatar’s chest movements with the user’s breathing enhances body ownership and agency (Monti et al. 2020), and mapping respiratory signals onto the expansion and contraction of a virtual avatar bubble viewed from a third-person perspective has been found to enhance agency and body awareness (Wald et al. 2023). What these object-oriented designs share is that respiratory feedback is concentrated in a single visual target, whether an external creature or a self-representation, which the user attends to as an indicator of their breathing. These designs are typically evaluated in terms of embodiment, measuring whether the temporal coupling between breathing and visual response enhances body ownership, felt agency, and awareness of the respiratory signal (Monti et al. 2020; Wald et al. 2023).

We contrast these with designs that embed respiratory feedback in the surrounding virtual environment rather than in a discrete object. DEEP (Bossenbroek et al. 2020; van Rooij et al. 2016) and Flowborne (Rockstroh et al. 2021) use diaphragmatic breathing to drive locomotion through immersive landscapes, rewarding paced breathing with movement and environmental change (Bossenbroek et al. 2020; Rockstroh et al. 2021). In a seated variant, Blum et al. (2020) mapped exhalation onto colour changes in flowers, trees, and rocks, keeping the feedback environmental but locally anchored to scene objects. A further group of environmental designs couples not the breathing signal directly but a slower downstream variable, typically heart

139 rate variability, to ambient scene properties. In [Blum et al. \(2019\)](#) and *Heart Garden*
 140 ([Rook et al. 2026](#)), increasing coherence or heart rate variability gradually transforms
 141 the scene. These designs create calming feedback loops, but because the environment
 142 changes through a slower physiological proxy rather than respiration itself, moment-
 143 to-moment respiratory agency is attenuated. The locomotion-based designs provide
 144 immediate agency, but mainly as a progression mechanic rather than as a way of mak-
 145 ing surrounding space itself feel breath-coupled. In our design, we sought to synthesize
 146 these two approaches by mapping the user’s breathing state onto an environment that
 147 expands and contracts with respiration. This provides immediate feedback and a sense
 148 of agency while presenting breathing as an ambient property of the surrounding space
 149 rather than as a distinct object requiring continuous attention. Specifically, the envi-
 150 ronment is composed of particles that respond to the user’s breath such that inhalation
 151 expands the space outward and exhalation contracts it inward. In this way, changes
 152 in internal lung volume are mapped onto changes in external space. This mapping
 153 is closely related to what [Goral et al. \(2024\)](#) describe as exteroceptive–interoceptive
 154 sensory substitution, in which internal bodily signals are translated into external sen-
 155 sory events that remain experientially linked to the body rather than appearing as
 156 separate readouts. By mirroring the felt expansion and contraction of the thorax,
 157 our system was intended to preserve the natural structure of breathing experience
 158 and thereby function as an intuitive interface that can be understood with minimal
 159 explicit instruction ([Antle et al. 2009](#)). This choice was also informed by evidence
 160 from embodied cognition showing that people judge quantities as larger after inhala-
 161 tion than after exhalation ([Belli et al. 2021](#)), which may indicate a possible, though
 162 still speculative, link between respiratory state and the perception of space. More
 163 broadly, this embodied metaphor appears across a range of immersive biofeedback
 164 systems that map respiration onto expanding and contracting visual stimuli ([Wald
 165 et al. 2023](#); [Goral et al. 2024](#); [Barton et al. 2024](#); [Stepanova et al. 2020](#)). Two exist-
 166 ing systems are conceptually closest to the approach presented here. [Barton et al.
 167 \(2024\)](#) immersed participants in an abstract fractal-like environment and used a cen-
 168 tral breathing guide whose size participants controlled by spreading and closing their
 169 arms in time with the guide; the interaction was therefore structured as an explicit
 170 movement-synchronisation task rather than as an environment that responded directly
 171 to respiration. [Goral et al. \(2024\)](#) introduced exteroceptive–interoceptive sensory sub-
 172 stitution in a system that translated respiratory signals into dynamic visual and
 173 auditory stimuli intended to feel continuous with the breathing body; however, their
 174 implementation externalized the signal through projection rather than embedding it in
 175 an immersive three-dimensional environment. Our system was designed to combine the
 176 strengths of these two approaches by pairing real-time environmental responsiveness
 177 that affords immediate respiratory agency with feedback distributed across surround-
 178 ing space rather than concentrated in a single object or task. Rather than relying on
 179 semantic cues such as naturalistic scenery, skyboxes, or rendered objects, we pursued
 180 a bottom-up design in which the sense of surrounding space emerges from relations
 181 among many simple particles, drawing on Gestalt principles such as proximity, com-
 182 mon fate, and continuity ([Wagemans et al. 2012](#)) so that their dynamic arrangement,
 183
 184

and specifically their responses to expansion and contraction, conveys depth and closeness dynamically within an abstract particle-based environment. This design rationale also motivates the broader question of whether coupling an internal bodily signal such as respiration to the surrounding sense of space can also reshape the sense of self, which we consider next through a review of relevant VR experiments.

1.3 VR interventions testing the malleability of the self

Virtual reality has been used extensively to investigate the malleability of bodily self-experience through multisensory manipulation. In full-body illusion paradigms, synchronous visuotactile stimulation between a participant’s physical body and a seen virtual body can shift perceived self-location and induce a sense of ownership over the virtual body (Lenggenhager et al. 2007; Petkova and Ehrsson 2008). Later work showed that interoceptive signals can serve a similar synchronising function. Presenting a virtual body or silhouette that flashes in synchrony with the participant’s heartbeat enhances body ownership and alters self-location even in the absence of tactile stimulation (Aspell et al. 2013; Suzuki et al. 2013). Respiratory signals have been used in a similar way. For example, synchronising a virtual avatar’s chest movements, viewed from a first-person perspective, with the user’s breathing enhances body ownership (Monti et al. 2020). Similarly, mapping respiratory signals onto the expansion and contraction of a virtual bubble, viewed from a third-person perspective, enhances agency and body awareness (Wald et al. 2023). Collectively, these studies show that a range of sensory modalities, including touch, breathing, and cardiac activity, can be used to reinforce visual feedback and thereby strengthen the sense of embodiment toward a virtual object or avatar from both first-person and third-person perspectives (Lenggenhager et al. 2007; Petkova and Ehrsson 2008; Aspell et al. 2013; Suzuki et al. 2013; Monti et al. 2020; Wald et al. 2023).

A related but distinct line of research concerns peripersonal space, the region immediately surrounding the body where the brain integrates multisensory signals in body-centered coordinates to maintain a dynamic boundary between self and environment (Serino 2019; Bufacchi and Iannetti 2018). This boundary is not fixed; it extends with tool use (Maravita and Iriki 2004), shifts with emotional valence and social context (Bufacchi and Iannetti 2018; Serino 2019), and interacts with interoceptive aspects of bodily self-representation (Tsakiris et al. 2011; Azzalini et al. 2019). Recent virtual-reality work further suggests that embodiment and multisensory manipulations can reshape peripersonal space in VR (Petrizzo et al. 2024; Karakoc et al. 2025). While the embodiment paradigms reviewed above manipulate the felt relationship between self and a visual body representation, peripersonal space research highlights that self-experience also depends on how the brain represents the spatial relationship between the body and the surrounding environment (Serino 2019; Bufacchi and Iannetti 2018).

Notably absent from both lines of research are paradigms that intervene through the surrounding environment rather than through a representation of the body itself. In most VR studies of embodiment and self-location, self-related experience is altered by manipulating how the body is represented, with any effects on perceived space arising indirectly through changes in how the body is seen in space (Guy et al. 2023). Although VR research has also developed techniques that manipulate space itself, such

231 as scaling, translation, and redirected movement, these are typically used to support
232 navigation and interaction rather than to investigate self-experience (Abtahi et al.
233 2022). By contrast, the present approach couples respiration not to a virtual body or
234 discrete object, but to the spatial extent of the surrounding environment itself. This
235 may engage the boundary between body-centred and environment-centred space in a
236 way that body-focused paradigms do not.

237 There is a broad consensus that the sense of self is multidimensional (Millière 2020;
238 Taves 2020), and the present study assessed three complementary pictographic mea-
239 sures. These were body-boundary salience (Dambrun 2016), spatial frame of reference
240 (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). At the same
241 time, the intervention was expected to affect not only self-experience, but potentially
242 the broader structure of subjective experience. The present study therefore also con-
243 sidered its effects in relation to psychometric frameworks from altered-states research,
244 which informed both measurement and the design rationale of the VR system described
245 next.

246

247 1.4 Breathwork as an experiential design space for virtual 248 reality 249

250 Embedding a breathing protocol in a responsive virtual environment introduces spe-
251 cific design constraints. Different breathing practices produce different respiratory
252 patterns, including variations in the pace, duration, and force of inhalation and
253 exhalation. Visual feedback coupled to these dynamics must therefore avoid becom-
254 ing dizzying, claustrophobic, disorienting, or overwhelming. In VR, large-amplitude
255 radial scene motion, particularly in the visual periphery, is a documented source of
256 cybersickness (Chang et al. 2020; Pöhlmann et al. 2021), and contracting environ-
257 ments can elicit involuntary defensive postural responses even in non-phobic users
258 (Kodithuwakku Arachchige et al. 2022). These risks are compounded when the user is
259 in a heightened or unfamiliar experiential state, given that negative affect and arousal
260 are associated with increased VR sickness severity (Kaufeld et al. 2022). The choice
261 of breathing protocol therefore shapes not only what physiological and experiential
262 effects to expect, but also what the visual environment can safely and comfortably do
263 in response.

264 High-ventilation breathwork practices such as holotropic breathing, conscious con-
265 nected breathing, and cyclic hyperventilation involve sustained increases in breathing
266 rate and depth that, over extended sessions, induce pronounced physiological changes,
267 including respiratory alkalosis and reductions in cerebral blood flow, and can give rise
268 to marked changes in subjective experience (Lewis-Healey et al. 2024; Kartar et al.
269 2025; Fincham et al. 2024). Participants report experiences of oceanic boundlessness,
270 in which the felt boundary between self and world dissolves into a sense of unity or
271 merging with the surroundings; visionary restructuralisation, involving vivid geomet-
272 ric imagery, complex visual scenes, and synesthetic blending of sensory modalities;
273 and dread of ego dissolution, characterised by a felt loss of control over one’s thoughts
274 and sense of identity (Bahi et al. 2024; Havenith et al. 2025; Lewis-Healey et al. 2024).
275 These experiential dimensions overlap with those reported under medium- to high-
276 dose psilocybin as measured by the 11D-ASC (Studerus et al. 2010; Havenith et al.

2025), and are often accompanied by emotional catharsis and strong bodily sensations such as tingling, dizziness, and autonomic arousal (Havenith et al. 2025; Lewis-Healey et al. 2024; Fincham et al. 2024, 2026). However, the rapid, large-amplitude chest excursions characteristic of high-ventilation protocols would produce correspondingly rapid expansion and contraction of a breath-coupled particle environment, and the accompanying physiological arousal and altered experiential state would coincide with precisely the visual conditions most likely to provoke cybersickness and spatial discomfort. For these reasons, we chose to facilitate slow-paced breathing rather than high-ventilation breathwork in the present system.

Slow-paced breathing occupies a different region of this subjective and affective space. Its established effect profile is characterised by relaxation, reduced anxiety, increased comfort, and calm alertness, supported by parasympathetic dominance and enhanced heart rate variability (Zaccaro et al. 2018; Balban et al. 2023; Fincham et al. 2023). Slow breathing is also one of the most common attentional anchors in guided meditation practice (Gerritsen and Band 2018; Zaccaro et al. 2018), and evidence from experienced pranayama practitioners indicates that very slow nasal breathing can induce altered states of consciousness characterised by altered body perception, distortions in time sense, unusual salience, increased joy, and reduced anxiety (Zaccaro et al. 2022). The mild but documented non-ordinary experiential profile of slow-paced breathing provided a low baseline against which any augmentation by the visual environment would be easier to detect, while keeping the environment’s spatial dynamics within a comfortable perceptual range.

Augmenting subjective experience is clinically relevant because experiential intensity during altered-state interventions predicts therapeutic outcomes (Roseman et al. 2018; Romeo et al. 2025; Havenith et al. 2025). We chose a slow-paced protocol precisely because its mild baseline provides sensitivity for detecting whether we can enhance specific attributes of the subjective experience through VR. To characterise this experiential profile and compare it with existing breathwork and psychedelic literatures, we adopted the 11D-ASC (Studerus et al. 2010; Dittrich 1998), the standard instrument across these fields. Its subscales capture the phenomenological features (e.g., boundary dissolution, changed imagery, synesthetic blending) that a breath-coupled environment might selectively amplify, and can be summarised into three higher-order dimensions reflecting positive, distressing, and perceptual effects (Stocker et al. 2025). Because we also examine how these dimensions evolve over the course of a breathing session, we include psychometric time-series ratings, which we refer to as temporal experience tracers (Jachs 2021). We match the tracers one-to-one with the 11D-ASC dimensions to compare global ratings with time-resolved intensity profiles of the same experiential constructs. The motivation to enhance subjective experience is also reflected in the aesthetic design of the visual environment, which we describe next.

1.5 Particle dynamics as a bioresponsive design medium

Viscereality is a bioresponsive VR system developed through an iterative design process, parts of which have been reported previously with respect to its design rationale, technical implementation, and interaction mechanics in human-computer interaction

contexts (Fejer et al. 2025, 2026a). The present study provides its first controlled empirical evaluation, testing whether the system’s particle dynamics measurably shape subjective experience during breathwork. As described above, the system maps lung volume onto a surrounding particle field through interoceptive-exteroceptive sensory substitution (Goral et al. 2024). Inhalation expands the environment outward, whereas exhalation contracts it inward. Existing implementations of this type of coupling typically use modelled surfaces, either presented as localized targets linked to the user (Wald et al. 2023) or projected onto the walls of a physical room (Goral et al. 2024). In both cases, the user watches a representation change size, but a uniform surface that grows or shrinks does not, on its own, strongly convey that the surrounding space itself is approaching or receding; that impression depends on motion-based depth cues that arise when many elements move relative to one another (Rogers and Graham 1979). We designed Viscerality as a particle system specifically to afford this. The environment is composed of approximately 2,500 particles distributed on a geodesic icosphere, and the sense of spatial expansion and contraction arises from Gestalt grouping among these elements, specifically proximity, common fate, and continuity (Wagemans et al. 2012), rather than from a texture mapped onto a surface. Because each particle independently carries multiple visual properties (position, size, colour, transparency, motion trajectory), the same elements that produce the breath-coupled spatial dynamics can simultaneously encode additional biofeedback channels, such as heart rate variability coherence (Fejer et al. 2026a), without requiring separate visual layers. And because inter-particle interaction rules, specifically oscillator coupling across neighbours, generate emergent aesthetic properties such as dynamic symmetry and geometric coherence, these can be parameterised independently of the breathing mechanics, making the aesthetic texture of the environment an experimentally controllable variable. Throughout the breathing cycle, the particle field remains continuously active. Surface-wave oscillations, chromatic cycles, and shape fluctuations continue during inhalations, exhalations, and both breath-hold phases. The within-VR manipulation therefore does not switch visual motion on or off; rather, during breath holds the high-symmetry condition transiently aligns these ongoing particle cycles into coherent geometric organisations, whereas the asymmetric condition leaves the same dynamics desynchronised. To our knowledge, no existing interoceptive-exteroceptive sensory substitution system takes this particle-based approach. Aesthetics in VR are often treated as incidental to intervention design, yet recent work identifies them as among the strongest predictors of emotional response and experience quality in immersive environments (Marcolin et al. 2021; Cheiran et al. 2025). In Viscerality, aesthetics are not a cosmetic layer but emerge from the same particle dynamics that carry the respiratory coupling, drawing on a longer history of expressive and aesthetic design in particle-based systems applied in visual effects animation.

1.5.1 Particle systems as a generative medium

Particle systems were introduced by Reeves (1983) to represent visual phenomena that have no fixed boundary and no stable shape, including fire, smoke, clouds, and water. In his formulation, each particle is born with stochastically assigned properties (position, velocity, colour, size, transparency, lifetime), moves and changes over time,

and eventually vanishes. The overall form is never fully specified; it emerges from the aggregate of many simple elements, each governed by a small set of controllable parameters. This is the property that makes particle systems useful as a bioresponsive medium because each particle independently carries multiple visual channels, so a single population of elements can simultaneously respond to physiological signals through spatial position, colour shifts, size, transparency, or motion trajectory without requiring separate visual layers for each mapping. Building on this work, Reynolds (1987) introduced interaction between elements, enabling coordinated collective behaviour to emerge from local neighbour rules, whereas in a standard particle system each element responds only to global parameters and physics without awareness of its neighbours. Reynolds specified three local rules. They were to avoid collision with nearby elements, match their velocity, and maintain proximity to them. This became foundational for the ability of 3D animation to produce the fluid, coordinated group motion recognisable in flocking birds, schooling fish, or stampeding herds, all arising bottom-up from neighbour interactions without a centrally scripted choreography.

1.5.2 Particle systems as an affective medium

Research on abstract emotion representation shows that affective meaning can be conveyed through low-level perceptual parameters of abstract expressions (de Rooij et al. 2013). In particular, affective valence can be mapped through semantically minimal features such as motion speed and trajectory smoothness (Bartram and Nakatani 2009; Pollick et al. 2001; Burger and Toiviainen 2020), angular or rounded shape contours (Blazhenkova and Kumar 2018; Aronoff 2006), and line orientation (Damiano et al. 2023). In conceptual terms, Khatin-Zadeh et al. (2019) argue that motion provides an effective vehicle for representing emotion because both unfold dynamically over time, allowing changes in affective state to be expressed through movement, direction, and position in space. This claim has also been borne out empirically in swarm robotics, where the collective motion of distributed abstract disc-shaped agents has been shown to alter the observer’s state of flow and sense of time relative to observing a single moving agent (Kaduk et al. 2024). Dietz et al. (2017) found that higher speed in swarm motion increased perceived emotional intensity, while smooth trajectories, relative to jittery motion, were associated with more positive affect. Extending this approach, Santos and Egerstedt (2021) translated trajectory and motion into swarm choreographies for Ekman’s six basic emotions (Ekman 1993). Rounded and smooth configurations marked positive or low-arousal states, while angular, clustered, or dispersive formations with fast or slow trajectories marked negative and higher-arousal states. Participants identified the intended emotions above chance (Santos and Egerstedt 2021). What makes these findings from swarm robotics relevant here is that their expressive effects arise from relational motion among many simple elements rather than from the form of any individual agent. The same principle holds in *Viscerality*, where local coupling among particles governs trajectory speed, smoothness, synchrony, clustering, and dispersal, producing higher-order collective motion patterns from which affective qualities may emerge. The particle field can therefore be treated as a medium for conveying affective states through abstract visual dynamics rather than semantic or figurative cues.

1.5.3 Particle systems as an aesthetic medium

Beyond serving as a medium for affective mapping, we designed the particle field as an aesthetic means of rendering affective states through abstract spatiotemporal organization. Research on abstract visual art suggests that non-figurative forms evoke aesthetic and affective responses through the isolation of simple features, the grouping of many elements into emergent wholes, and the amplification of perceptual qualities through Gestalt grouping principles (de Rooij et al. 2013; Ramachandran and Hirstein 1999). In our particle system, these same grouping principles also help convey depth when approximately 2,500 ring-shaped particles expand and contract around the user. Because each particle is visually minimal, higher-order forms are not pre-given but emerge through local relations among neighbouring elements, allowing circles, lattices, polygons, and other geometric organisations to arise through changes in three-dimensional location. The appearance of such forms can itself become aesthetically salient, as the resolution of an initially indeterminate field into a coherent Gestalt is known to produce pleasurable moments of insight or ‘aesthetic Aha’ moments (Muth and Carbon 2013; Spehar and van Tonder 2017; Spee et al. 2024). In Viscerality, this made it possible to translate oscillator-level control into changes in overall visual order, symmetry, and coherence without moving to figurative imagery.

Another design influence was the geometric and kaleidoscopic qualities of hallucinatory and psychedelic imagery, which are often marked by geometric patterns, bilateral symmetry, and perceptual ambiguity (Klüver 1966; Billock and Tsou 2012; Lawrence et al. 2022; Rubin et al. 2010). This motivated our use of ring-shaped particles, as their overlaps can give rise to distinctive emergent patterns during the breathing cycle. Because these qualities emerge from shifting relations among many simple particles rather than from fixed symbolic forms, they remain open in meaning and foreground a defining affordance of the medium, namely its capacity to continually reorganise and transform its aesthetic structure over time.

In the context of VR design, Glowacki (2024) terms such non-figurative scenes a weak representational aesthetic, in contrast to photorealistic depictions that render scenes and bodies as recognizable real-world entities. In practice, this aesthetic has been applied in VR scenarios designed to induce altered states of consciousness through the representation of participants’ bodies as luminous, semi-corporeal avatar bodies composed of particle clouds with mergeable body boundaries, contributing to the malleability of self-experience and altered subjective states (Glowacki et al. 2020, 2022; Glowacki 2024; Vidal et al. 2025). Our approach adopts a related design aesthetic while focusing more specifically on diminishing the felt boundary between self and environment.

To evaluate the particle system’s capacity to manipulate aesthetic properties, we operationalized symmetry as an additional independent variable that varied across conditions. Previous studies have linked visual symmetry to more positive affect and perceptual fluency (Bertamini et al. 2013; Pecchinenda et al. 2014; Makin et al. 2012), while synchrony in group movement has been linked to greater enjoyment and more positive affective appraisal (Vicary et al. 2017; Mogan et al. 2017). We therefore asked whether higher degrees of symmetry would enhance positively valenced experiences

in the context of Viscerality (Johnson 2023). We were also interested in whether participants would be able to detect and discriminate between the two conditions.

1.6 Present Study

The present study provides the first controlled empirical evaluation of Viscerality with naive participants. We used a preregistered within-subjects design with $N = 39$. Each participant completed three 10-minute audio-guided box-breathing sessions in counterbalanced order. These were a high-symmetry condition, in which positive oscillator coupling transiently aligned ongoing particle oscillations into dynamic geometric coherence during breath-hold phases; an anti-symmetry condition, in which negative coupling produced a visually irregular particle field throughout; and a dark-screen control (see Section 2.3.4). The two particle conditions used identical particles, breath-coupled spatial dynamics, colour palette, and audio guidance, differing only in oscillator coupling parameters. The central within-VR contrast therefore concerned the aesthetic organisation of an otherwise identical breath-responsive scene.

We assessed four substantive outcome domains together with a methodological blinding assessment. Breathing adherence was quantified as the phase-locking value between the observed respiratory signal and an ideal reference trajectory. Self-related experience was assessed with three pictographic measures, namely body-boundary salience (Dambrun 2016), spatial frame of reference (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). Altered states of consciousness were assessed retrospectively with the 11D-ASC (Studerus et al. 2010; Dittrich 1998). Blinding efficacy between the two active VR conditions was assessed retrospectively with a two-stage procedure. Before unmasking, participants completed 12 probe questions covering condition-specific, shared, and sham features. After being told that the two active scenes differed, they indicated in which order the high-symmetry and anti-symmetry scenes had been presented, together with confidence ratings. These responses were used to estimate condition-identification blinding with a standard single-group Bang blinding index and an exploratory confidence-weighted directional variant (Bang et al. 2004).

We hypothesised that, relative to the dark-screen control, the bioresponsive VR conditions would increase breathing adherence and altered-state intensity and would also influence the three preregistered self-related measures. Within VR, we further expected the high-symmetry condition to enhance positively valenced experiences and produce stronger alterations in self-related experience than the anti-symmetry condition. The preregistration did not specify a separate directional prediction for each pictographic outcome. We therefore treated the three outcomes as one family and controlled the false-discovery rate across them.

2 Methods

2.1 Participants

Participants were recruited from the University of Konstanz psychology student participant pool and received course credit for participation. The study used a within-subject

507 design in which each participant completed all three conditions in one of six coun-
508 terbalanced block-order permutations, with a preregistered target of 6 completers per
509 permutation and 36 total. We did not systematically assess prior experience with
510 virtual reality, meditation, breathwork, or psychedelics. We use recruitment-naïve to
511 mean that participants came from the general psychology participant pool through an
512 advertisement for a VR breathing-biofeedback study rather than an altered-state or
513 experience-seeking recruitment channel; it does not mean that prior experience was
514 screened or absent. We tested 47 participants; 8 were excluded based on breathing
515 signal quality (phase-locking value (PLV) outlier criterion described below), yielding
516 a final sample of $N = 39$ (6–7 per permutation), preserving approximately equal rep-
517 resentation across all six condition sequences. Among the included participants, mean
518 age was 20.8 years; 37 were female (mean age 20.7 years) and 2 were male (mean age
519 23.0 years).

520

521 2.2 Design, Conditions, and Procedure

522

523 Participants were seated in a bean bag, wore noise-blocking headphones, held a con-
524 troller against their abdomen, and completed all breathing blocks, questionnaires,
525 and outcome measures fully immersed in virtual reality. The three-session procedure
526 was automated end-to-end in the headset: once the headset was put on, participants
527 followed in-VR instructions for the breathing blocks and all post-block outcome mea-
528 sures while remaining seated, with minimal experimenter intervention beyond initial
529 setup and debriefing. Before the experimental sequence, participants completed a
530 1-minute guided breathing practice block for acclimation, familiarizing them with con-
531 troller placement and the paced-breathing rhythm. Each participant then completed
532 three conditions, with session order assigned automatically by random selection from
533 the six counterbalanced block-order permutations. Because the dynamic symmetry
534 manipulation was embedded in the software-controlled scene logic and the condi-
535 tion sequence was assigned automatically at runtime, neither the participant nor the
536 experimenter knew which condition would appear in which session. In all conditions,
537 participants followed a 10-minute audio-guided box-breathing exercise (sourced from
538 a publicly available recording) consisting of repeated 4-4-4-4 cycles with four seconds
539 each of inhalation, breath hold, exhalation, and breath hold. In the high-symmetry
540 condition, the bioresponsive particle environment formed dynamic symmetric patterns
541 during breath-hold phases. In the asymmetric condition, visual features of the particles
542 remained asymmetric to one another throughout. In the dark-screen control condition,
543 the particle environment was not displayed; participants viewed a dark field contain-
544 ing only a central fixation marker that also provided breathing-performance feedback
545 (Section 2.3.4). We refer to this condition as the “dark-screen control” throughout.
546 After each block, participants completed the full assessment battery (questionnaires
547 and pictographic scales) while remaining in virtual reality. Figure 1 summarizes the
548 full procedure, including the repeated condition loop and blinding sequence, using the
549 same condition-color mapping as the Results figures.

550

551

552

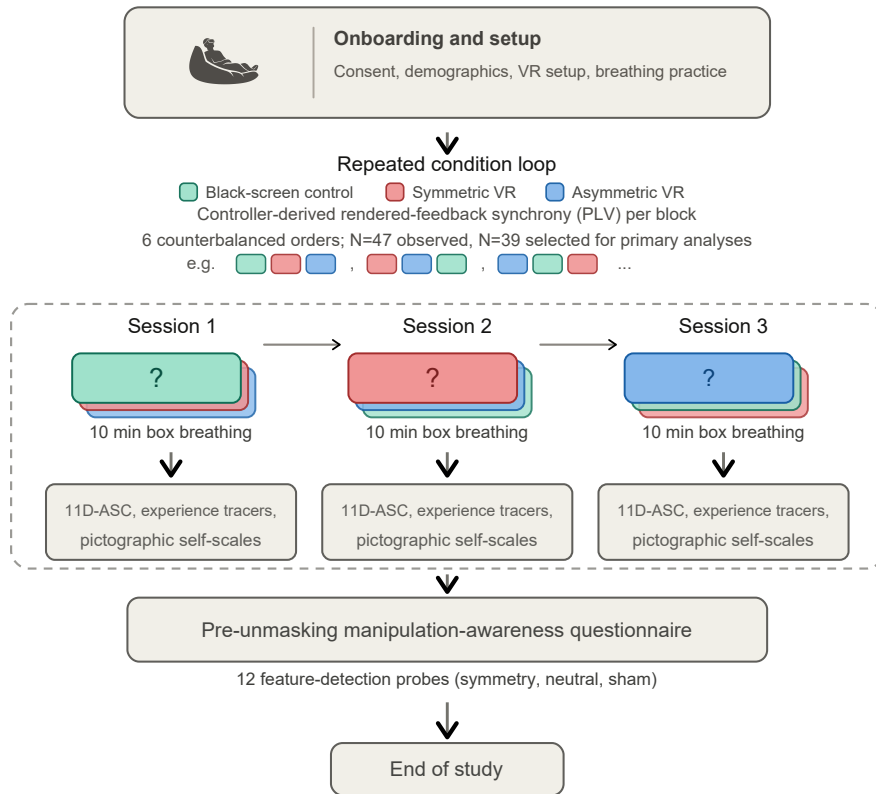


Fig. 1 Study procedure. After onboarding, each participant completed three 10-minute box-breathing sessions (dark-screen control, symmetric VR, asymmetric VR) in one of six counterbalanced orders, with the full assessment battery administered in VR after each block. A blinding questionnaire followed the third block before the study concluded

2.3 Virtual-Reality Intervention

Viscereality is a bioresponsive virtual-reality system that maps respiratory signals to dynamic visual feedback within an immersive 3D particle environment (Fejer et al. 2025, 2026a). A video summary of the system is available online.¹ The system runs on a Meta Quest 3 headset (standalone, untethered) and was built in Unity, with CPU-optimized parallel processing (Unity Job System with Burst compilation) maintaining a stable 72 Hz frame rate. The two active virtual-reality conditions used the same visual environment, audio guidance, and logging pipeline; they differed only in a single engine parameter controlling oscillator coupling dynamics. The dark-screen control used identical audio guidance and data logging but omitted the particle visualization. We focus here on the experiment-relevant functional logic; fuller technical

¹<https://www.youtube.com/watch?v=Qm-lmC-xYCI>; archived at <https://osf.io/4enfz>

599 and implementation details of the Viscereality engine are described in the prior system
600 papers (Fejer et al. 2025, 2026a).

601

602 2.3.1 Breath Detection

603

604 Breath detection follows the approach introduced by Flowborne (Rockstroh et al.
605 2021), using the VR controller’s positional tracking to infer respiratory displacement
606 without additional sensors. During an initial calibration, the user places the controller
607 against the lower abdomen to establish a tracking axis. Each frame, the algorithm reads
608 the controller’s position and orientation, projects motion onto this breathing-relevant
609 axis, and updates a short-term versus long-term smoothed motion estimate. The dif-
610 ference between these smoothed signals functions as a breathing trend signal. Positive
611 values indicate inhalation, negative values indicate exhalation, and near-zero values
612 indicate a pause. Samples are rejected as tracking artifacts if controller rotation is too
613 abrupt or if the inferred trend is implausibly large. The resulting classification (inhale,
614 exhale, pause, or bad tracking) drives the visual feedback pipeline described below
615 (Figure 2). Detailed implementation choices and parameterization are documented in
616 the Viscereality technical descriptions (Fejer et al. 2025, 2026a).

617

618 2.3.2 Sphere Radius and Visual Feedback

619

620 The user is positioned at the center of a large particle-covered sphere. The primary
621 feedback loop couples sphere radius to breathing. Inhalation causes the sphere to
622 expand outward, exhalation causes it to contract inward, and breath holds main-
623 tain the current radius. Functionally, the controller-derived breathing-state label is
624 converted into a signed expansion/contraction command whose magnitude is scaled
625 to the configured radius range and the intended inhale/exhale timing. That com-
626 mand updates a target sphere radius, and the rendered radius is then smoothed with
627 a damped interpolation step (0.25 s smooth time; overshoot prevention enabled). In
628 practice, this means participants do not see a raw controller-motion trace; they see a
629 bounded, temporally stable visual signal that preserves breathing direction while sup-
630 pressing jitter. Full implementation details of the radius controller are described in
631 the prior Viscereality system reports (Fejer et al. 2025, 2026a).

632

633

634

635

636

637

638

639

640

641

642

643

644

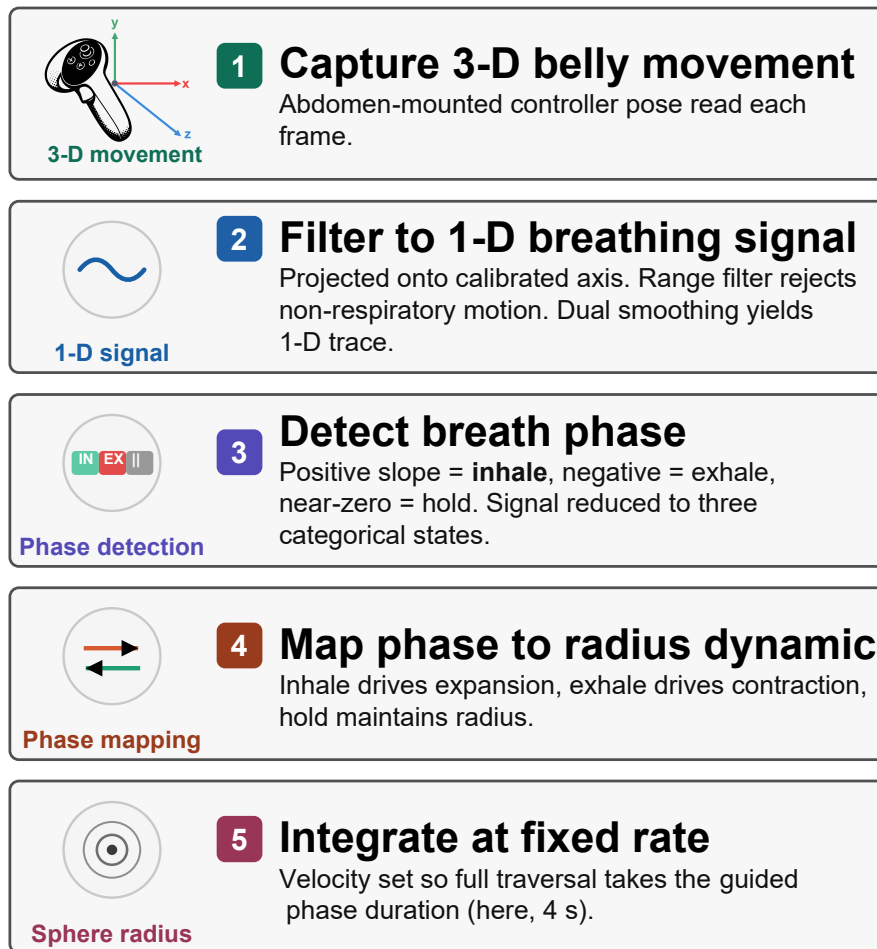


Fig. 2 Signal processing pipeline from abdomen-mounted controller motion to rendered sphere radius. Raw 3D movement is projected onto a calibrated breathing axis, classified into inhale, exhale, or hold phases, and mapped onto expansion and contraction dynamics that update the visible sphere at a fixed velocity

The sphere surface is composed of approximately 2,500 view-aligned ring particles distributed on an icosphere mesh for uniform geodesic spacing. Each particle is rendered with a radial gradient that produces depth cues through overlap, and during expansion and contraction, particles generate cone-like tracer effects that reinforce the perception of three-dimensional movement. Surface wave oscillations, structured noise, and unsynchronized chromatic cycles maintain continuous visual dynamics across all breathing phases, preventing the environment from appearing static during breath holds.

2.3.3 Oscillator Network and Condition Manipulation

The second layer of the system implements a Kuramoto oscillator network across the sphere’s vertices (Acebrón et al. 2005; Rodrigues et al. 2016). Each particle has an internal oscillation phase (analogous to a clock angle) and interacts with its neighbors through coupling weights that determine whether nearby particles tend toward synchronization or away from it. If the coupling weight is positive, neighbors are pulled toward shared timing (phase attraction). If negative, the interaction becomes *anti-coupling*, and neighbors are pushed away from alignment (phase repulsion).

The network uses a two-tier neighbor structure reflecting the icosphere topology. Most vertices have 6 immediate (tier-1) neighbors and 12 next-nearest (tier-2) neighbors, while 12 vertices at the original icosahedron poles have 5 tier-1 neighbors. At each update step, each oscillator advances according to its own intrinsic rhythm plus a neighborhood interaction term. In plain terms, the algorithm compares an oscillator’s current phase with those of its neighbors and then adjusts it toward or away from them depending on the sign and magnitude of the tier-specific coupling weights. This is the standard Kuramoto logic (phase-offset-dependent attraction/repulsion), implemented here with two neighbor tiers and a global coupling gain, and updated in parallel each frame together with particle position, size, and color. The formal implementation and parameterization are described in the prior Viscerality publications (Fejer et al. 2025, 2026a).

The two active conditions differed in how coupling weights were assigned. In the high-symmetry condition, tier-1 and tier-2 coupling weights were interpolated dynamically as a function of a coherence scalar (0–1), together with a coherence-dependent frequency scaling term. Verbally, low coherence pushes the system toward anti-coupled, irregular dynamics, whereas higher coherence progressively shifts interactions toward phase attraction and more synchronized motion. As a result, particle motion, color, and size periodically coordinate into coherent geometric patterns, such as flower-of-life-like lattices, during breath-hold phases before dissolving back into disorder when breathing resumes.

In the asymmetric condition, the system did not use this dynamic interpolation. Instead, coupling weights were held at fixed negative values (anti-coupling), so local interactions were permanently biased toward phase repulsion and the particle field remained visually irregular throughout. The manipulation was thus dynamical rather than geometric. The same scene, particles, and audio were used in both conditions, and only the oscillator interaction rules changed.

2.3.4 Breathing Performance Indicator

All three conditions displayed a minimal breathing performance indicator at the center of the participant’s field of view. This consisted of a white pentagon ring whose individual line segments scaled with sphere radius: at half-volume (midpoint between full inhale and full exhale), the pentagon was fully visible; as the participant inhaled or exhaled toward the extremes, the segments progressively shortened and disappeared entirely at maximum and minimum volume. The indicator thus functioned as an unobtrusive loading-bar-style monitor of breathing excursion. Participants were instructed

that the goal was to make the pentagon disappear completely on each inhalation and exhalation, confirming sufficient breathing depth.

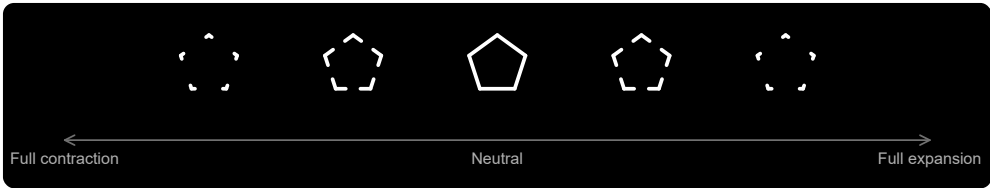


Fig. 3 Central breathing-performance indicator shown in all three conditions. The pentagon ring was fully visible at the neutral sphere radius and progressively disappeared toward full contraction or full expansion, providing condition-matched feedback about breathing excursion

2.4 Outcome Measures

We retained the five preregistered measurement blocks used in the assessment battery, but organize them here into four recurring outcome families that also structure the Results section. These were adherence to box breathing, self-related measures, altered states of consciousness, and the blinding questionnaire. Within the altered-states family, the main text presents the derived 3D-ASCr composites as a dimensionality-reduced summary; Online Resource 1, Section S3 reports the complete preregistered 11D-ASC analyses.

2.4.1 Adherence to Box Breathing

Adherence quantifies how closely each participant’s breathing followed the instructed 4-4-4-4 box-breathing rhythm across each 10-minute block. The preregistered plan was to compute this from discrete phase labels produced by the breath-detection algorithm (Section 2.3.1), which classifies each frame as inhale, exhale, hold, or bad tracking. The intended metric was phase-duration error, defined as the absolute deviation from the 4-second target per breathing phase, averaged across valid cycles per condition, with participant exclusion if fewer than 70% of cycles were valid or if mean error exceeded 2 standard deviations above the sample mean.

The recorded phase sequences did not yield the expected clean succession of 4-second phases. Rather than discrete blocks of inhale, hold, exhale, and hold, the recorded data consisted of highly fragmented segments interspersed with brief transitional or ambiguous intervals. Computing phase-duration error from this data required first reconstructing plausible box-breathing cycles from the raw phase stream, a step that was not anticipated at preregistration. We explored several reconstruction strategies, including rule-based merging of adjacent micro-segments and temporal smoothing algorithms, but each approach introduced its own arbitrary parameter choices, such as minimum segment duration thresholds, smoothing window widths, and rules for combining adjacent same-phase fragments, that could materially influence the resulting error estimates. Because these post-hoc binning decisions carried more researcher degrees of freedom than the original preregistered metric implied, we adopted an alternative approach.

We quantified adherence as the phase-locking value (PLV) between two continuous time series. One was the sphere-radius signal that participants actually saw as visual feedback during the breathing session. The other was an ideal reference radius trajectory derived from the audio-instruction timestamps for perfect 4-4-4-4 breathing. PLV measures how consistently the phase relationship between two signals remains stable over time. A value near 1 indicates that the participant’s breathing closely tracked the instructed rhythm throughout the block. A value near 0 indicates little consistent alignment. This approach circumvents the phase-reconstruction problem entirely, because it evaluates alignment at the level of the continuous visual feedback signal rather than attempting to recover discrete phase boundaries from a fragmented controller trace. PLV thus captures the same underlying construct as the preregistered metric, namely how closely breathing tracked the instructed rhythm, while avoiding the measurement artifacts that any post-hoc phase-segmentation strategy would have introduced. Figure 4 illustrates the relationship between observed and reference trajectories for a representative participant; the fragmented phase-label structure visible in the lower bars of each panel motivates the PLV-based approach described above.

Because PLV does not decompose into per-cycle validity counts, the preregistered 70%-valid-cycle plus 2-SD mean-error exclusion rule could not be transferred directly. Outlier exclusion instead used a lower-tail global IQR rule on condition-level PLV values, elevated to participant-level exclusion if any condition was flagged. Online Resource 1, Section S2 describes PLV computation and quality screening in detail.

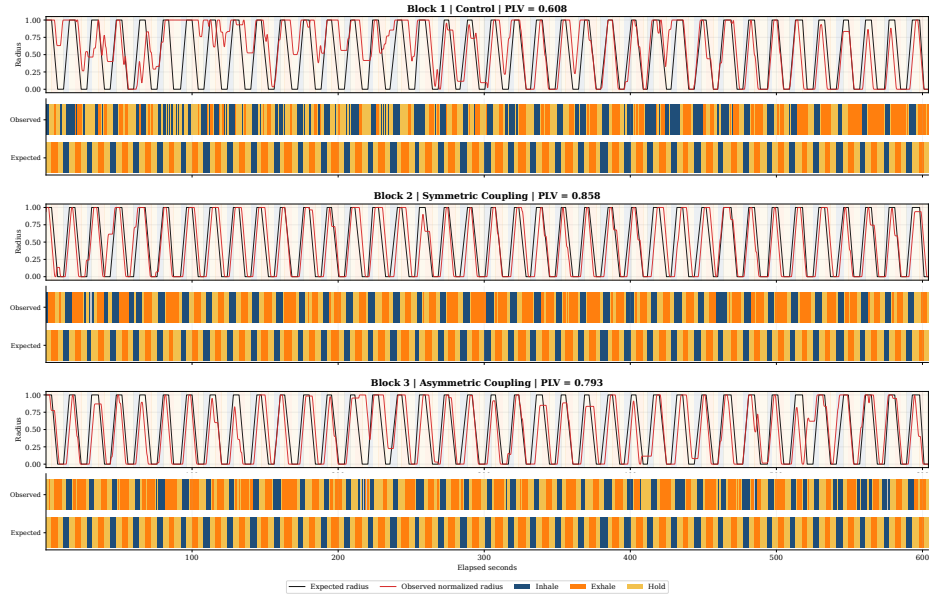


Fig. 4 Breathing dynamics for a representative participant (P01) across all three conditions. Observed sphere-radius traces (orange) are overlaid on the ideal 4-4-4-4 reference (blue), with per-block PLV values shown at right. Fragmented phase-label bars below each time series illustrate the segmentation noise that motivated the switch from phase-duration error to PLV as the adherence metric. Equivalent figures for all participants are provided in the OSF repository

2.4.2 Self-Related Measures

Three single-item pictographic measures assessed self-related experience after each condition. The adapted Perceived Body Boundaries item (Dambrun 2016) presents seven silhouettes progressing from diffuse (A) to sharply delineated (G). Responses were mapped from A–G to 1–7, so higher analyzed scores indicate stronger boundary salience. For construct-facing exploratory correlations, the item was reverse-scored so that higher values indicate greater boundary dissolution. The Spatial Frame of Reference Continuum (SFoRC; Hanley et al. 2020) depicts a figure in concentric spatial zones from body-centered to environment-encompassing, indexing the degree to which the spatial sense of self extended beyond the body. The Small Self pictographic item was adapted from Vidal et al. (2025), who presented it as an ad hoc item. We included it because Viscerality is organized around a spatial metaphor and the item represents self-related experience through the spatial size of the self relative to its surroundings. PBBS and Small Self use A–G responses mapped to 1–7, whereas SFoRC uses A–H responses mapped to 1–8. Each measure yields a single ordinal score per condition. We recreated all three pictographs as SVG graphics for VR presentation, using the previously published versions as visual references. The vector format allowed direct and flexible control over graded visual properties, including transparency, line width, dash spacing, relative figure size, and spatial extent, while enabling sharp rendering at headset resolution without the resolution loss associated with scaling raster images. We generated the SVG files in December 2025 with OpenAI Codex using GPT-5.2-Codex, based on author instructions and the cited published pictographs. The authors reviewed the resulting SVG files and take responsibility for the final stimuli. The OSF reproducibility package (<https://osf.io/64mwk/>) provides the recreated stimulus set, SVG source files, administered item keys, and study materials; Online Resource 1, Section S6 provides the corresponding reproducibility map.

2.4.3 Altered States of Consciousness

3D-ASCr composites. The revised 3D-ASCr higher-order factor model became available after preregistration. It provides a parsimonious contemporary summary of the same preregistered 11D-ASC responses by organizing them into Positive, Distressing, and Perceptual Effects. This level of aggregation matches our broad theoretical interest in positively versus negatively valenced experience, while retaining a perceptual dimension, and avoids overinterpreting lower-order dimensions for which no specific predictions were made. We therefore use the 3D-ASCr composites as the main-text summary. Online Resource 1, Section S3 reports complete analyses of all 11 preregistered 11D-ASC dimensions.

11D-ASC Questionnaire. We assessed altered states with the 11-Dimensional Altered States of Consciousness Rating Scale (11D-ASC; 42 items, 0–100 visual analogue scales), a validated lower-order factorization of the OAV/APZ framework (Studerus et al. 2010; Dittrich 1998; Hovmand et al. 2024; Prugger et al. 2022). The 11 subscales were Experience of Unity (EU), Spiritual Experience (SE), Blissful State (BS), Insightfulness (IS), Disembodiment (DIS), Impaired Control and Cognition

875 (ICC), Anxiety (ANX), Complex Imagery (CI), Elementary Imagery (EI), Audio-
876 Visual Synesthesia (AVS), and Changed Meaning of Percepts (CMP). Participants
877 completed one condition-specific rating after each block.

878
879
880
881
882
883
884
885
886
887
888
889
890
891
892
893
894
895
896
897
898
899
900
901
902
903
904
905
906
907
908
909
910
911
912
913
914
915
916
917
918
919
920

Temporal Experience Tracers (Exploratory). Temporal experience tracers provide psychometric time-series ratings of the same 11 experiential dimensions.

For each dimension, one prompt consolidated the constituent questionnaire items into a construct-level description. After each condition, participants retrospectively drew the 11 intensity curves on a world-space time-by-intensity surface using the VR controller. Controller position was projected onto the drawing plane, input was gated through a trigger hold, and points were emitted at constant spatial spacing regardless of hand speed. Each trace was reduced to peak intensity, representing the strongest reported moment, and area under the curve (AUC), representing intensity accumulated across reconstructed session time (Figure 5). We focused on these two summaries because no hypothesis required finer temporal features such as onset, timing, slope, or variability. We treat the tracer analyses as exploratory; Online Resource 1, Section S5 provides the complete implementation, data reduction, analyses, and results.

**Temporal Trace Profiles:
Peak versus Area Under the Curve**

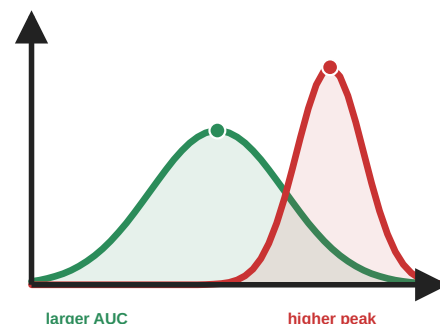


Fig. 5 Temporal experience tracers: peak and cumulative intensity. These hypothetical profiles illustrate information retained by psychometric time-series intensity ratings but not by a single retrospective visual-analogue-scale response. The red profile has a higher momentary peak, whereas the broader green profile has a larger area under the curve (AUC), reflecting greater cumulative intensity. Analyses in this paper focus on these two summaries because no hypothesis required finer-grained temporal features

2.4.4 Blinding Questionnaire

Blinding was assessed in two phases. In the pre-unmasking phase, participants were told that the study investigated how subtle differences in visual aesthetics might influence perception in virtual reality. We constructed a 12-item ad hoc masked-comparison questionnaire for this study. For each item, participants chose whether the feature was stronger in Scenario 1, stronger in Scenario 2, equal in both scenarios, or absent in both scenarios. The chance baseline was 0.25. Scenario labels referred to the two virtual-reality conditions in encounter order. The item set covered three hidden domains. These were manipulation-relevant symmetry/coordination cues that genuinely differed between the two VR conditions, matched comparison features that were present similarly in both conditions and therefore should be judged as equal, and foil (sham) features that were not present in either condition and therefore should be judged as absent. There were four items per domain.

In shorthand, the four symmetry-related items asked whether one scene more strongly showed transient coordination and ordering of the particle field, including particles moving as a coordinated group, becoming more organized or structured at certain moments, showing wave-like coordinated motion-and-color changes, or shifting from chaotic to ordered movement during breath holds. The four neutral matched

967 items asked about features that were present in both VR conditions to a similar degree,
968 namely particles becoming brighter during breath holds, leaving faint motion trails
969 during exhalation, remaining in gentle ongoing motion across the surface, or appearing
970 to rise and fall like ocean waves. The four sham items were constructed as absent cross-
971 modal or pulsing foils, asking about the music becoming quieter, the music seeming to
972 slow down, particle motion and color changing in response to the music, or particles
973 showing a size-based pulsing effect. To keep administration uniform while avoiding an
974 obviously grouped structure, items from all three domains were intermixed in a sin-
975 gle randomized order that was generated once in advance and then presented in that
976 same order to all participants.

977 In the post-unmasking phase, participants were told that one condition featured
978 brief visual synchronization during breath holds while the other remained irregular
979 throughout. They indicated which session was more symmetric via a three-option
980 forced choice (*first*, *second*, or *I don't know*), followed by a confidence rating.

981

982 2.5 Preregistration and Reproducibility

983

984 The study was preregistered on AsPredicted (#262,545). The unchanged registry
985 record, all deidentified participant-level data, statistical analysis scripts and out-
986 puts, figure-generation scripts, and study materials were uploaded to OSF as a
987 reproducibility package (<https://doi.org/10.17605/OSF.IO/64MWK>). We retained all
988 preregistered design elements and outcome families. The preregistered analysis plan
989 specified a repeated-measures ANOVA with condition as the within-subjects factor and
990 presentation order as a covariate for each outcome, two planned contrasts (pooled VR
991 versus dark-screen control, and symmetric versus asymmetric), and parallel Bayesian
992 analyses. Adherence was to be quantified as phase-duration error with a cycle-validity
993 exclusion rule, as described and motivated in Section 2.4.1. We deviated from this
994 plan in several respects. Online Resource 1, Section S1 summarizes the deviations from
995 preregistration and their rationale.

996 **Adherence metric.** Phase-duration error was replaced by PLV because the recorded
997 phase-label data were too fragmented to yield reliable per-phase error estimates with-
998 out introducing arbitrary binning decisions (see Section 2.4.1 for a full account). The
999 associated exclusion rule was changed from 70%-valid-cycle plus 2-SD mean-error to
1000 a lower-tail global IQR rule on condition-level PLV values, because PLV does not
1001 decompose into per-cycle validity counts.

1002 **Omnibus condition test.** For every non-PLV outcome, the preregistered order-
1003 adjusted repeated-measures analysis tests the condition effect after accounting for
1004 participant and numeric block position. It is implemented as a partial- F comparison
1005 between a participant-and-order model and the same model plus condition. Partici-
1006 pant fixed effects provide within-participant blocking, and block position adjusts for
1007 presentation order. The finalized PLV omnibus remains unchanged. A condition-only
1008 repeated-measures ANOVA without the order covariate is retained in the OSF outputs
1009 as a sensitivity check.

1010

1011 **Planned contrasts.** The two preregistered planned contrasts were implemented as
1012 standalone paired t -tests, which yield the same paired contrast statistics as ANOVA

follow-ups in this within-subject setting. These serve as the confirmatory anchor for all multi-test outcome families in the Results section, and for significant planned contrasts in the altered-state families we additionally report paired common-language effect sizes (CLES), defined as the proportion of participants whose score was higher in the VR condition than in the dark-screen control (McGraw and Wong 1992).

Robustness and multiplicity. Because Shapiro–Wilk checks indicated non-normality for several condition-level and difference-score distributions, we added Wilcoxon signed-rank and Friedman tests as secondary distribution-free checks. The preregistration did not specify a multiplicity procedure. Nominal p values therefore govern the primary interpretation of the preregistered planned contrasts, with Benjamini–Hochberg q values reported as sensitivity information within the 11D-ASC, 3D-ASC_r, and self-related outcome families. FDR-adjusted q values govern exploratory and post hoc screens. The OSF package contains the complete adjusted and distribution-free output; the paper reports material divergences from the primary planned-contrast interpretation.

3D-ASC_r composites. The revised 3D-ASC_r scoring scheme (Stocker et al. 2025) became available after preregistration. Because these composites are deterministic averages of the preregistered 11D-ASC subscale scores, they add no new participant data but do constitute a post-registration summary choice. We therefore use them to organize the main-text narrative while retaining the complete preregistered 11D-ASC analyses in Online Resource 1, Section S3.

Bayesian analyses. Bayesian analyses were preregistered and run in parallel on the same included sample ($N = 39$). Omnibus Bayes factors compared a point-null model without condition (H_0 : outcome $\sim$ order + participant random intercept) against a model that additionally included condition (H_1 : outcome $\sim$ condition + order + participant random intercept), using the BayesFactor R package with explicit prior settings (fixed-effect $r = 0.50$, continuous-covariate $r = \sqrt{2}/4 \approx 0.354$, random-effect nuisance $r = 1.00$; Morey and Rouder 2024; van Doorn et al. 2021; Kruschke 2021). For the two planned paired contrasts, we compared H_0 ($\delta = 0$) against two-sided H_1 ($\delta \neq 0$) using a Bayesian paired t -test with the default Cauchy prior on standardized effect size ($r = \sqrt{2}/2 \approx 0.707$; Morey and Rouder 2024; van Doorn et al. 2021). For EU, BS, and AVS, the preregistered joint ordering symmetric > asymmetric > dark-screen control was tested after block-position adjustment with an encompassing-prior Bayes factor against the equal-means null. Under the exchangeable condition-factor prior, each of the six strict condition orderings had prior probability 1/6. We report BF_{10} values (and $BF_{01} = 1/BF_{10}$ when the data favor the null) as relative evidence rather than as posterior probabilities or effect sizes (van Doorn et al. 2021; Kruschke 2021). Posterior median standardized effect sizes with 95% credible intervals and prior-robustness checks are reported in the OSF repository.

Blinding analyses. Pre-unmasking accuracy was scored using the documented answer key for the four symmetry, four neutral, and four sham items and across all 12 items. Mean accuracy was compared with the 0.25 chance baseline using Wilcoxon signed-rank tests, with Benjamini–Hochberg FDR correction across the four comparisons. Post-unmasking guesses were classified as correct (R), incorrect

(W), or uncertain (U) from the participant’s selected active-VR session and condition order. The standard single-group Bang blinding index was calculated as $BBI = (R - W)/(R + W + U)$ (Bang et al. 2004; Kolahi et al. 2009), with a participant-bootstrap 95% confidence interval. Effective blinding was considered established only if the entire interval fell within $[-0.2, 0.2]$ (Qin et al. 2024; Kolahi et al. 2009). We also report an exploratory confidence-weighted directional index, calculated as the confidence-weighted proportion of correct directional guesses minus 0.5. The standard Bang interpretation bands were not applied to this modified index. To test whether pre-unmasking symmetry accuracy predicted subsequent condition identification, we calculated a point-biserial correlation between the 0–4 symmetry score and final-guess correctness, classified as correct versus incorrect or uncertain, with an exact two-sided permutation p value.

Exploratory analyses. We fit condition $\times$ PLV interaction models to test whether breathing synchronization moderated subjective effects. We also conducted an exploratory analysis of whether VR-related changes in the pictographic measures covaried with changes in ASC outcomes. For focused, construct-based interpretation, we examined SFoRC in relation to Experience of Unity and Disembodiment, including items concerning unity with the environment, floating, bodylessness, and altered self-location outside the body. We examined perceived body-boundary dissolution primarily in relation to Disembodiment and the same bodily-self items, and Small Self in relation to Experience of Unity, Spiritual Experience, and Blissful State. These thematic selections were post hoc and guided interpretation rather than defining separate inferential families. We additionally conducted broad exploratory screens across all 3D-ASCr composites, 11D-ASC subscales, and ASC items. Because only SFoRC showed a reliable pooled-VR-versus-dark-screen-control condition effect, the correlations were treated as exploratory tests of cross-measure correspondence rather than as evidence that all three pictographic measures changed under VR. Online Resource 1, Section S4 reports the complete analysis, with false-discovery-rate correction across all 9 composite, 33 subscale, and 126 item-level associations, together with symmetric-minus-asymmetric change correlations and order-adjusted participant-aware condition-interaction models.

3 Results

Across the four recurring outcome families, condition effects were consistently driven by the contrast between virtual-reality conditions and the dark-screen control. No outcome showed a reliable symmetric-versus-asymmetric difference. We report the results in the same family order used in Methods: adherence, self-related measures, altered states of consciousness, and the blinding questionnaire. For the non-PLV multi-test families, confirmatory interpretations are anchored in the preregistered planned contrasts evaluated as paired t -tests, with the registered order-adjusted participant-blocked partial- F omnibus noted for each family as context; PLV retains its finalized mixed-effects omnibus. Wilcoxon signed-rank robustness checks, Friedman omnibus tests, and FDR-adjusted q values are documented in the OSF repository and are mentioned here only when they materially qualify the paired t pattern.

3.1 Sample and Exclusions

The dataset contained records from all 47 tested participants. Condition-level PLV values from all participants were used to apply the quality-screening procedure described in Section 2.4.1 and illustrated in Figure 6. This procedure excluded 8 participants and yielded a final analytic sample of $N = 39$. The 39 included participants each contributed a complete within-subject triad and constituted the final sample used for all inferential analyses.

3.2 Outcome Families

3.2.1 Adherence (PLV)

To test whether visual biofeedback improved adherence to the guided box-breathing rhythm, we compared PLV across conditions. Mean PLV was $M = 0.787$ ($SD = 0.141$) for the dark-screen control, $M = 0.836$ ($SD = 0.127$) for symmetric VR, and $M = 0.860$ ($SD = 0.094$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed this difference, $t(38) = 3.47$, $p = .001$, $d_z = 0.56$, 95% CI for the mean PLV difference $[0.026, 0.097]$. The symmetric-versus-asymmetric contrast was not significant, $t(38) = -1.13$, $p = .267$, $d_z = -0.18$. The order-adjusted mixed-effects omnibus supported a condition effect, $\chi^2(2) = 12.66$, $p = .002$. The corresponding Bayesian planned contrasts showed that the pooled-VR-versus-dark-screen-control difference was about 24.1 times more likely than no difference ($BF_{10} = 24.11$), whereas the symmetric-versus-asymmetric data were about 3.2 times more likely under a no-difference model ($BF_{01} = 3.22$).

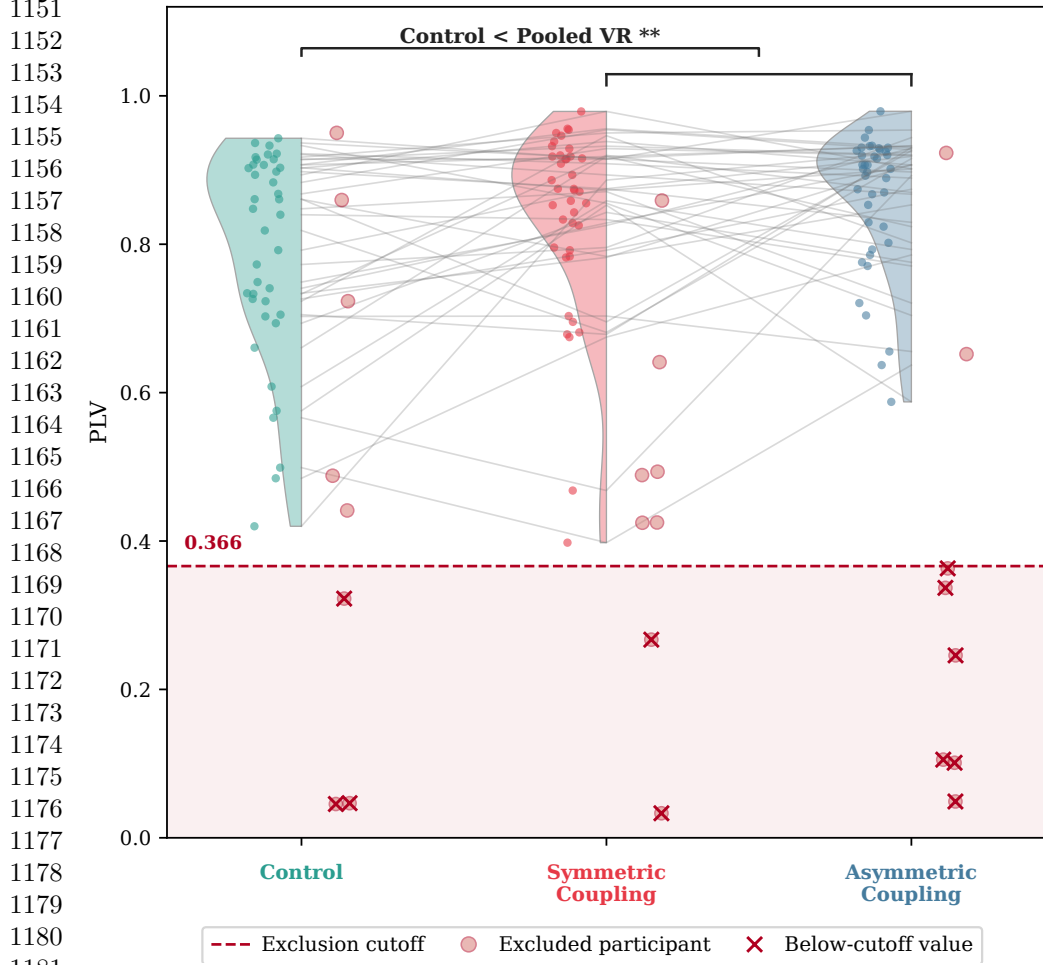


Fig. 6 Controller-derived feedback synchrony (PLV) by condition for the selected sample ($N = 39$), with condition-wise distributions and paired individual trajectories in gray. The dashed red line marks the pooled lower-tail IQR exclusion cutoff (0.366); hollow circles and red crosses indicate flagged participants and their below-cutoff values, respectively. If any single condition fell below the cutoff, the participant was excluded entirely from all analyses across all conditions

3.2.2 Self-Related Measures

To assess whether the breath-responsive virtual environment shifted subjective self-representation, we compared three pictographic measures across conditions. Only the spatial frame-of-reference continuum showed a condition effect. Participants reported a more spatially extended sense of self during virtual-reality conditions, with $M = 3.15$ ($SD = 1.41$) for the dark-screen control, $M = 4.18$ ($SD = 1.82$) for symmetric VR, and $M = 4.26$ ($SD = 1.73$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed the effect, $t(38) = 4.28$, $p < .001$, $d_z = 0.69$, 95% CI [0.56, 1.57]. The symmetric-versus-asymmetric contrast was not

significant, $t(38) = -0.32$, $p = .752$. The registered order-adjusted omnibus result is $F(2, 75) = 10.99$, $p < .001$, $q_{FDR} < .001$, $\eta_p^2 = .227$. Neither perceived body-boundary salience ($t(38) = -0.17$, $p = .864$) nor small self ($t(38) = 1.39$, $p = .173$) showed a significant VR-versus-dark-screen-control contrast, and no self-related measure showed a significant symmetric-versus-asymmetric contrast. Figure 7 shows the response formats together with the condition-wise distributions and paired participant trajectories. Online Resource 1, Section S4 provides the complete registered frequentist and Bayesian condition-analysis inventory for all three pictographic measures.

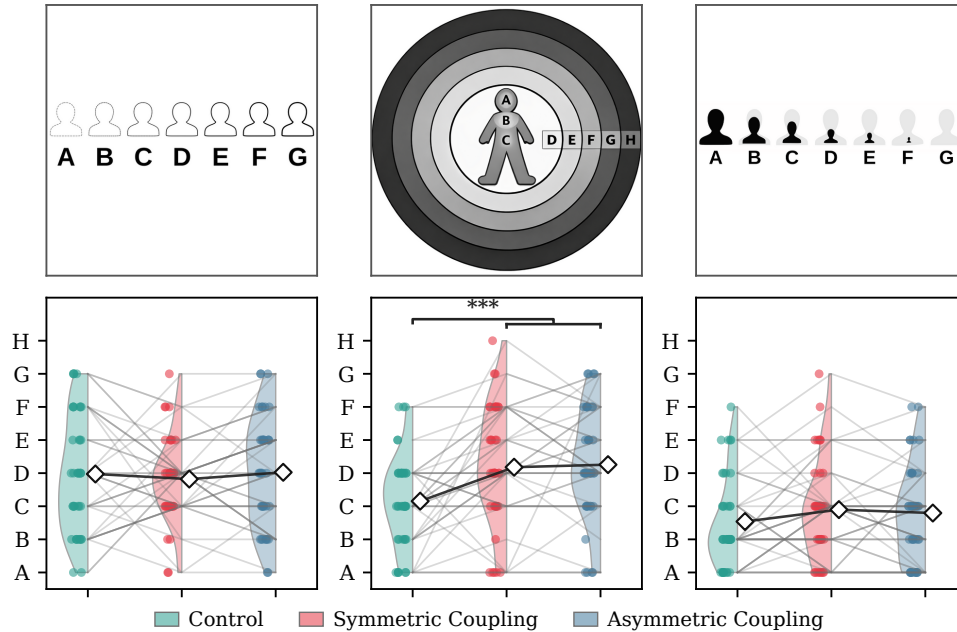


Fig. 7 Condition-wise distributions for the three pictographic self-related measures ($N = 39$). The upper row shows the administered response formats, and the lower row shows condition-wise distributions, paired participant trajectories, and means (diamonds), from left to right: Perceived Body Boundaries, the Spatial Frame of Reference Continuum (SFoRC), and Small Self. Perceived Body Boundaries is displayed in its administered A–G direction, from diffuse to sharply delineated, so higher scores indicate stronger boundary salience. Only SFoRC showed a reliable pooled-VR-versus-dark-screen-control difference. Perceived Body Boundaries and Small Self showed no reliable pooled-VR-versus-dark-screen-control differences, and no pictographic measure differed reliably between the symmetric and asymmetric VR conditions. Colors indicate the dark-screen control, symmetric VR, and asymmetric VR. The SVG pictographs in the upper row were generated in December 2025 with OpenAI Codex using GPT-5.2-Codex, based on author instructions and the published pictographs from the Perceived Body Boundaries Scale (Dambrun 2016), the Spatial Frame of Reference Continuum (Hanley et al. 2020), and the Small Self item (Vidal et al. 2025); the authors reviewed the resulting SVG files

Bayesian comparisons agreed with this pattern at the planned-contrast level. For spatial frame of reference, the observed data were approximately 204 times more likely under a pooled-VR-versus-dark-screen-control difference model than under the

no-difference model ($BF_{10} = 204.44$). Perceived body boundaries favored the no-difference model ($BF_{01} = 5.71$), whereas the Small Self result provided only weak and inconclusive evidence in the null direction ($BF_{01} = 2.39$).

Exploratory pictograph-ASC change-score analyses are reported in Online Resource 1, Section S4. Their clearest pattern was coordinated participant-level variation between spatial self-extension and positive or unity-related experience, but these post hoc associations do not alter the primary condition-effect conclusions.

3.2.3 Altered States of Consciousness

We report the altered-state family in the main text using the 3D-ASCr composites as a dimensionality-reduced summary. Online Resource 1, Section S3 reports the complete preregistered 11D-ASC analyses. The clearest pattern is stronger positive and perceptual VR-related effects, with a weaker distress-related component and no reliable symmetric-versus-asymmetric difference.

3D-ASCr composites.

Summarizing the 11D-ASC data through the 3D-ASCr composites (see Section 2.4.3 for derivation), the three higher-order dimensions showed the following pattern. Perceptual Effects had $M = 11.41$ ($SD = 12.61$) in the dark-screen control, $M = 27.25$ ($SD = 19.92$) in symmetric VR, and $M = 25.59$ ($SD = 17.24$) in asymmetric VR. Positive Effects had $M = 18.88$ ($SD = 15.39$) in the dark-screen control, $M = 25.64$ ($SD = 19.48$) in symmetric VR, and $M = 26.46$ ($SD = 17.84$) in asymmetric VR. Distressing Effects had $M = 12.33$ ($SD = 10.26$) in the dark-screen control, $M = 17.33$ ($SD = 12.66$) in symmetric VR, and $M = 17.02$ ($SD = 13.18$) in asymmetric VR. All three composites showed significant pooled-VR-versus-dark-screen-control planned contrasts: Perceptual Effects $t(38) = 4.90$, $p < .001$, $d_z = 0.79$; Positive Effects $t(38) = 3.22$, $p = .003$, $d_z = 0.51$; Distressing Effects $t(38) = 2.56$, $p = .014$, $d_z = 0.41$. Expressed on the native 0–100 scale, pooled VR increased Perceptual Effects by 15.0 points, Positive Effects by 7.2 points, and Distressing Effects by 4.8 points. These corresponded to a large effect for Perceptual Effects, a moderate effect for Positive Effects, and a smaller effect for Distressing Effects, with 76.9%, 69.2%, and 76.9% of participants, respectively, scoring higher in VR than in the dark-screen control.

No composite showed a symmetric-versus-asymmetric difference (all $p \geq .357$). The registered order-adjusted omnibus results are $F(2, 75) = 19.63$, $p < .001$, $q_{FDR} < .001$, $\eta_p^2 = .344$ for Perceptual Effects; $F(2, 75) = 7.56$, $p = .001$, $q_{FDR} = .002$, $\eta_p^2 = .168$ for Positive Effects; and $F(2, 75) = 4.64$, $p = .013$, $q_{FDR} = .013$, $\eta_p^2 = .110$ for Distressing Effects. Bayesian omnibus comparisons favored condition models for Perceptual Effects ($BF_{10} = 121,810$), Positive Effects ($BF_{10} = 30.49$), and, more modestly, Distressing Effects ($BF_{10} = 3.45$). The planned Bayesian contrasts showed the same hierarchy. For pooled VR versus control, the observed data were approximately 1,178 times more likely under a difference model for Perceptual Effects ($BF_{10} = 1,178.41$), 13 times more likely for Positive Effects ($BF_{10} = 12.96$), and only 3 times more likely for Distressing Effects ($BF_{10} = 3.02$). For symmetric versus asymmetric VR, the observed data favored the no-difference models by factors of approximately 3.9, 5.1, and 5.7, respectively ($BF_{01} = 3.87$, 5.07, and 5.66; Figure 8). Thus, the Perceptual and Positive

pooled-VR effects had clear relative support, whereas the smaller Distressing effect had only modest Bayesian support.

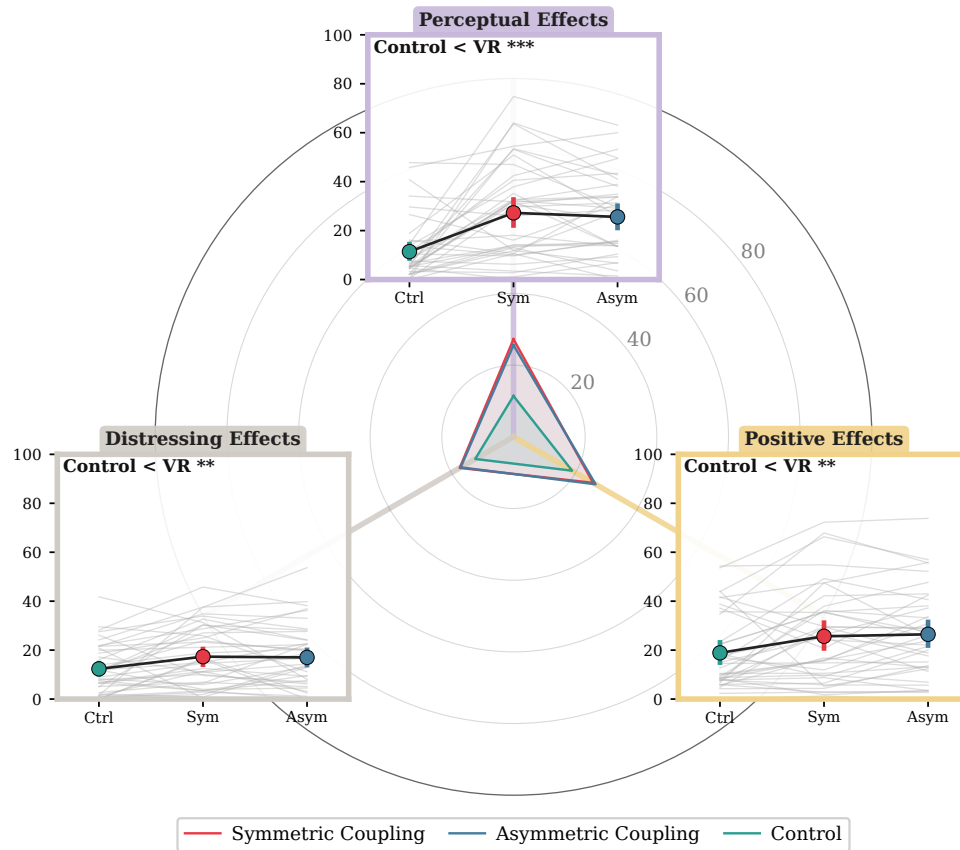


Fig. 8 Condition-wise distributions for the three 3D-ASCr composites ($N = 39$). All three dimensions increased in VR relative to the dark-screen control, with the largest effect for Perceptual Effects; no symmetric-versus-asymmetric differences emerged

11D-ASC questionnaire.

Complete condition descriptives and preregistered frequentist and Bayesian results for all 11 lower-order ASC dimensions, including the registered directional hypotheses, are reported in Online Resource 1, Section S3.

3.2.4 Blinding Questionnaire

Pre-unmasking responses showed selective but incomplete sensitivity to the symmetry manipulation. Mean symmetry-item accuracy was 44.2%, equivalent to approximately 1.77 of four items, and exceeded the 25% chance baseline (FDR-adjusted $q < .001$). Neutral-item accuracy was 27.6% ($q = .610$), and sham-item accuracy was 32.7%

1335 ($q = .100$); neither was clearly above chance. Across all 12 items, mean accuracy was
1336 34.8% ($q < .001$). Participants therefore detected some genuine symmetry features,
1337 but performance remained well below perfect identification.

1338 Post-unmasking guesses were nearly balanced: 17 participants guessed correctly
1339 (43.6%), 19 guessed incorrectly (48.7%), and 3 were uncertain (7.7%). The standard
1340 Bang blinding index was close to zero ($\text{BBI} = -.051$, 95% bootstrap CI $[-.359, .256]$).
1341 Confidence weighting changed little: the exploratory confidence-weighted index was
1342 $-.016$ (95% bootstrap CI $[-.203, .168]$, $n = 35$ directional responses). Both intervals
1343 crossed zero, indicating no reliable tendency toward either correct or opposite identifi-
1344 cation. However, the standard BBI interval extended beyond the predefined $[-0.2, 0.2]$
1345 blinding region. The results were therefore consistent with effective blinding but too
1346 uncertain to rule out meaningful identification in either direction.

1347 To test whether pre-unmasking sensitivity predicted subsequent condition iden-
1348 tification, we related the 0–4 symmetry score to final-guess correctness, classified as
1349 correct versus incorrect or uncertain. Pre-unmasking symmetry-detection accuracy
1350 was descriptively higher among participants who subsequently identified the correct
1351 condition (2.00 versus 1.59 of four items), but the association was small and incon-
1352 clusive (point-biserial $r = .18$, exact two-sided permutation $p = .33$; Figure 9d).
1353 Thus, better masked detection did not reliably predict the final condition guess in this
1354 sample.

1355
1356
1357
1358
1359
1360
1361
1362
1363
1364
1365
1366
1367
1368
1369
1370
1371
1372
1373
1374
1375
1376
1377
1378
1379
1380

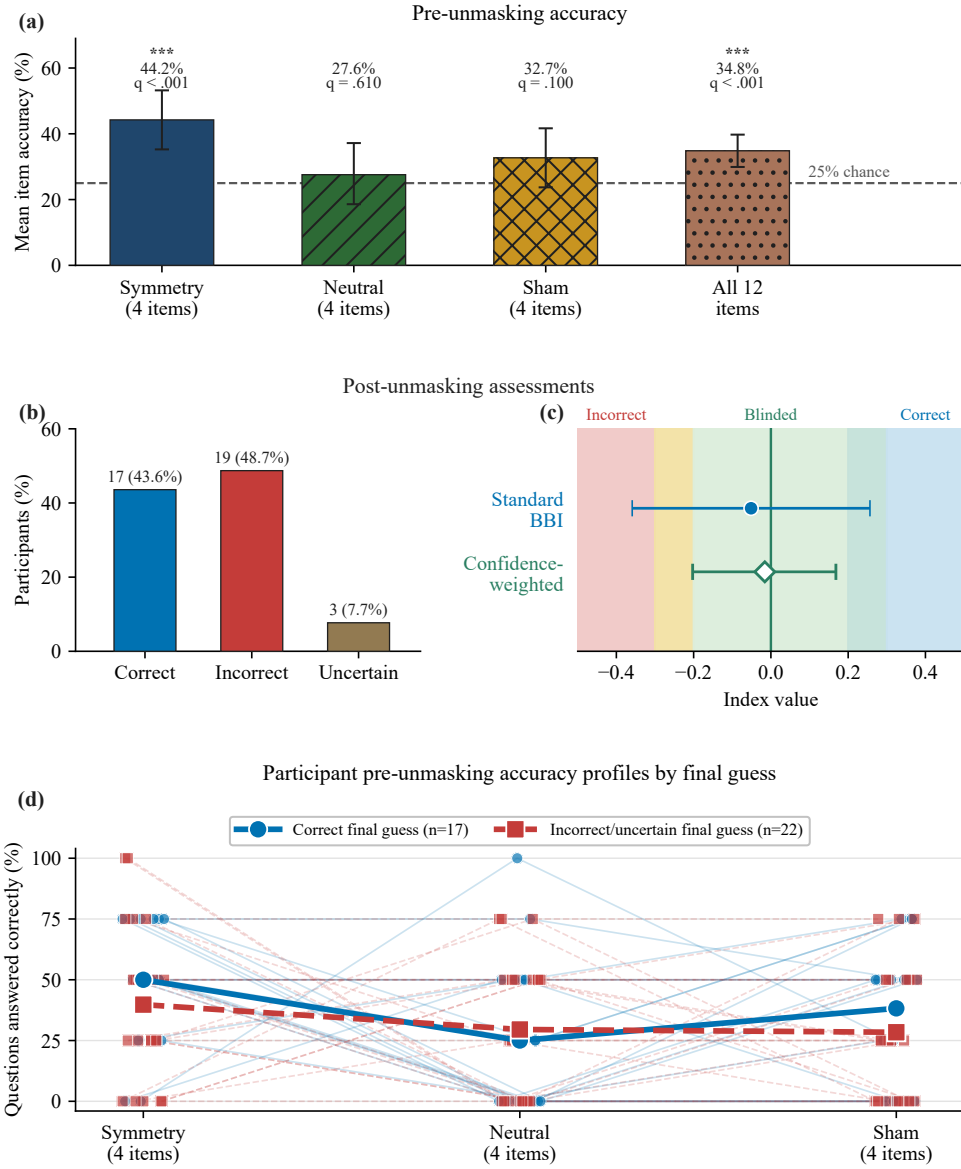


Fig. 9 Blinding and manipulation-awareness summary ($N = 39$). (a) Mean pre-unmasking accuracy for the four symmetry, four neutral, and four sham items and across all 12 items. The dashed line marks the 25% chance baseline; asterisks denote FDR-adjusted $q < .001$. (b) Post-unmasking guesses: 17 correct, 19 incorrect, and 3 uncertain. (c) Standard Bang blinding index (filled blue circle) and exploratory confidence-weighted directional index (open green diamond), with participant-bootstrap 95% confidence intervals. The shaded regions indicate incorrect, blinded, and correct identification and apply only to the standard Bang index. (d) Participant-level pre-unmasking accuracy across the three four-item categories, grouped by final-guess correctness. Thin lines and small markers represent participants; heavier lines and larger markers represent group means. Incorrect and uncertain guesses are combined

1427 3.3 Exploratory Checks

1428 Exploratory mixed-model checks for order and PLV moderation are reported in the
1429 OSF repository. No FDR-corrected order main effects, condition $\times$ order, condition $\times$
1430 PLV, or condition $\times$ order $\times$ PLV interactions emerged for any ASC, 3D-ASCr, or self-
1431 related outcome family. For PLV itself, there was a main effect of order, $\chi^2(1) = 8.28$,
1432 $p = .004$, but no condition $\times$ order interaction, $\chi^2(2) = 0.37$, $p = .833$, indicating
1433 that block order did not change the between-condition adherence pattern. A planned
1434 linear contrast across session position was significant, $t(38) = -3.16$, $p = .003$, 95% CI
1435 $[-0.100, -0.022]$, indicating that PLV decreased rather than increased across sessions
1436 and therefore aligning more with waning engagement or fatigue than with a practice-
1437 improves account.

1439 4 Discussion

1441 This study provided a first controlled empirical evaluation of Viscereality with naive
1442 participants. Relative to audio-guided box breathing in the dark-screen control, the
1443 two VR conditions improved breathing adherence, shifted spatial self-experience
1444 toward a more expanded frame of reference, and produced an altered-state profile in
1445 which perceptual and positive effects outweighed mild distressing effects. None of the
1446 47 tested participants discontinued the experiment because of discomfort or inability
1447 to tolerate the VR exposure. However, the preregistered aesthetic manipulation
1448 that varied the symmetry of the particle dynamics did not reliably differentiate the
1449 two active VR conditions. Taken together, these findings indicate that the paradigm
1450 produced a structured experiential profile during a brief breathing session with naive
1451 participants, but stronger control comparisons and objective measures are needed to
1452 determine the specificity and depth of these effects.

1454 4.1 Breathing adherence and self-related experience

1455 Participants in both active VR conditions followed the instructed breathing pattern
1456 more closely than in the dark-screen control. The display responded to participants'
1457 ongoing breathing rather than to the audio guide, so it made respiration continuously
1458 visible without telling participants how well they were matching the instructed pac-
1459 ing. One plausible interpretation is that the breath-responsive environment helped
1460 adherence by keeping participants engaged and by giving them the sense that their
1461 breathing was producing the visual experience. Previous findings are consistent with
1462 this. Breath-coupled VR feedback has been linked to improved focus on breathing and
1463 greater diaphragmatic movement (Blum et al. 2020; Rockstroh et al. 2021), greater
1464 focused attention and less distraction than non-VR biofeedback (Rockstroh et al.
1465 2019), and increased interoceptive sensibility (Goral et al. 2024). A simpler explana-
1466 tion also remains possible. Guided breathing in the dark-screen control is presumably
1467 less engaging than the same task in an immersive environment, and the adherence met-
1468 ric itself was derived from how closely breathing matched the trajectory expressed by
1469 the visual display. The study would have benefited from a more stringent control, such
1470 as a visually matched scene that does not respond to breathing or one that replays
1471

a prerecorded animation, because the engaging visual environment on its own may have contributed to the effect. This caveat matters because controlled comparisons in the VR breathing literature have not consistently shown VR-based interventions to outperform matched non-VR approaches on clinical outcomes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). The present adherence finding should therefore be treated as preliminary until tested against a more stringent control.

Of the three self-related measures, only the spatial frame-of-reference continuum showed a condition effect. We had reason to expect body-boundary salience to shift as well because prior work reports reduced perceived boundary salience during meditation (Dambrun 2016); in one study that administered both scales, mindfulness training reduced perceived body boundaries and expanded allocentric spatial framing in tandem, with the boundary change mediating the spatial shift (Hanley et al. 2020). That pattern did not emerge here, perhaps because the two constructs respond to partly different triggers. Body-boundary dissolution appears to shift most reliably during practices that direct sustained attention toward the body itself, such as body-scan meditation (Dambrun 2016), mindfulness training (Hanley et al. 2020), and floatation-REST (Hruby et al. 2024). Spatial frame of reference, by contrast, has been linked not only to body-directed contemplative practice (Hanley et al. 2020) but also to perceived expansion of the physical frame of reference beyond the body (Dambrun 2023). Because *Viscerality* acts on the surrounding visual-spatial environment rather than directing attention toward the body, it may have engaged the spatial-frame construct through this environmental route without producing the body-focused conditions under which boundary dissolution typically occurs. Of the three scales, spatial frame of reference is also the one most closely aligned with the broader question motivating this research programme, namely whether breath-coupled changes in surrounding space can alter the felt extension of self into the environment. Previous work has shown that mapping respiration onto an externalized virtual avatar can modulate body ownership (Monti et al. 2020) and self-location (Adler et al. 2014; Betka et al. 2020) through afferent respiratory pathways (Betka et al. 2020).

However, this result requires cautious interpretation. The concentric depiction of the SFoRC scale may have resembled the expanding and contracting particle sphere shown in VR, which raises the possibility that participants selected more spatially expansive depictions because the scale imagery matched what they had just seen rather than because their felt self-boundary had shifted. This demand-characteristics concern cannot be ruled out with pictographic self-report alone and represents a clear limitation of the present measurement approach. The finding should accordingly be treated as preliminary pending corroboration from measures that do not share visual features with the stimulus, such as peripersonal space tasks. If confirmed by such measures, the spatial-frame shift would support continued investigation of whether mapping respiration onto surrounding space can modulate the felt extension of self into the environment.

Within the context of a 10-minute box-breathing exercise with ambient background music, *Viscerality* increased all three higher-order dimensions of the Altered States of Consciousness Rating Scale relative to the dark-screen control, which provided only a central fixation marker and minimal visual breathing-performance feedback.

1519 Perceptual Effects showed the largest increase, Positive Effects a moderate increase,
1520 and Distressing Effects a smaller increase. None differed reliably between the two
1521 VR mappings that manipulated visual symmetry, and participants showed only lim-
1522 ited feature-level sensitivity to this manipulation without reliable condition-level
1523 identification.

1524 The fine-grained subscale decomposition and item-content analysis reveal a more
1525 specific experiential profile (Online Resource 1, Section S3.4). The perceptual shift
1526 primarily concerned simple patterns, colors, lights, and flashes, together with impres-
1527 sions that sound changed visual shapes or colors. These elementary visual phenomena
1528 closely matched features supplied directly by the particle environment, so their increase
1529 partly reflects overlap between the intervention and the questionnaire criteria. Par-
1530 ticipants also attributed visual changes to sound even though the same background
1531 music was presented in every condition and did not control the display. One plausible
1532 interpretation is that the richer and more dynamic visual stream provided more per-
1533 ceptual material that could be bound to, or retrospectively attributed to, concurrent
1534 sound, rather than producing audio-visual synaesthesia in the stricter sense. A similar
1535 distinction has been raised in psychedelic research, where LSD increased spontaneous
1536 synaesthesia-like reports that lacked the consistency and inducer specificity associated
1537 with genuine synaesthesia (Terhune et al. 2016).

1538 The Positive Effects increase was concentrated in items reflecting a loosening of
1539 separation between self, surroundings, and time, heightened significance and emotional
1540 salience of ordinary objects, sensations of floating or diminished bodily presence, and
1541 moments of awe.

1542 The smaller Distressing Effects increase had the character of diffuse destabilization,
1543 encompassing reduced agency, difficulty organizing thought, isolation, unexplained
1544 apprehension, and an altered or threatening quality of the surroundings. Neither
1545 constituent distress-related subscale remained reliable after correction across the fine-
1546 grained analyses. We therefore interpret the composite increase as an accumulation of
1547 mild and compatible unpleasant experiences rather than as a pronounced or clearly
1548 localized adverse response.

1549 Descriptively, the pooled-VR ASC profile was compared with five breathwork
1550 datasets: conscious connected breathing (Bahi et al. 2024), the 90-minute breathwork
1551 arm of Airways to Alteration (Fincham et al. 2026), facilitated circular breathwork
1552 (Havenith et al. 2025), high-ventilation breathwork with music in an MRI/laboratory
1553 setting (Kartar et al. 2025), and Holotropic Breathwork (Uthaug et al. 2021). The Vis-
1554 cereality profile fell within the range of several of these longer breathwork datasets and
1555 showed comparatively lower distress than facilitated circular breathwork. Differences
1556 in breathing technique, duration, sample, music, facilitation, setting, and control struc-
1557 ture make this comparison contextual rather than evidence of equivalence. Figure 10
1558 shows the full profile overlay, and Online Resource 1, Section S3 retains the fine-grained
1559 subscale commentary.

1560

1561

1562

1563

1564

Cross-Study ASC Profile Comparison

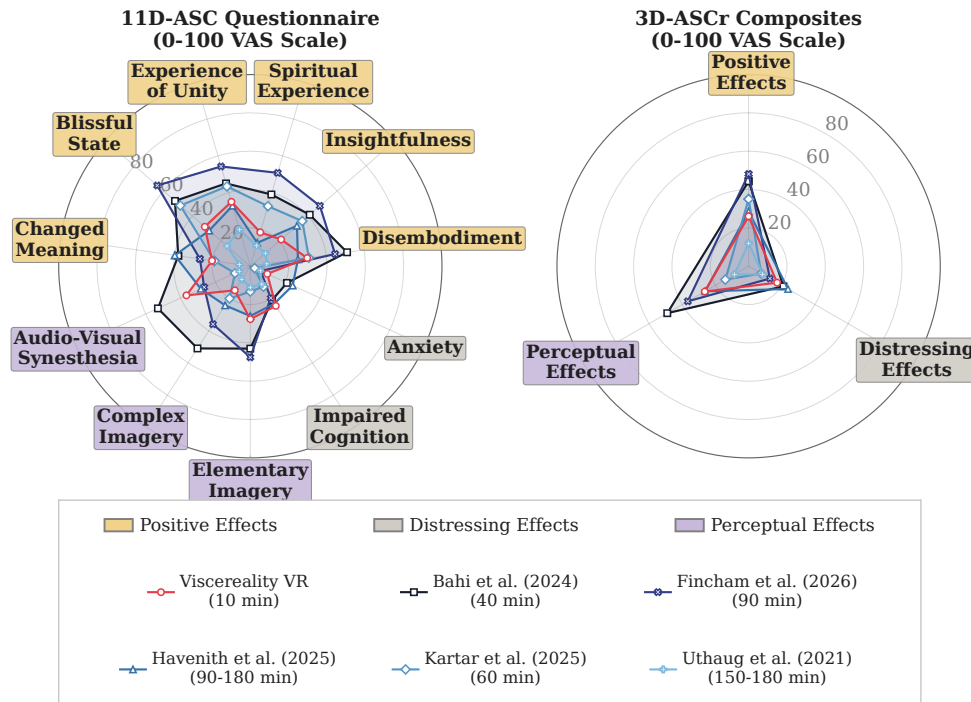


Fig. 10 Descriptive cross-study comparison of the present dataset with five named breathwork datasets. The left panel shows 11D-ASC subscale means and the right panel shows 3D-ASCr composite means. The Viscerality profile pools the symmetric and asymmetric VR conditions. Lines show raw 0–100 profile means, not standardized effect sizes; differences in technique, duration, sample, music, facilitation, and study design preclude inferential comparison. Legend order is purely graphical and does not imply a numbered study sequence

At subscale level, the profiles differed most in Experience of Unity, Spiritual Experience, Blissful State, Disembodiment, Insightfulness, and Changed Meaning of Percepts. Changed Meaning of Percepts increased in pooled VR relative to the dark-screen control in the present study, whereas Insightfulness did not change reliably. Because the reference studies differ on multiple procedural and contextual dimensions, these profile differences do not identify the mechanism responsible for any between-study contrast.

The blinding results distinguish masked feature-level sensitivity from post-unmasking condition identification. Before unmasking, symmetry-item accuracy exceeded chance, while neutral and sham accuracy did not. Participants therefore detected some genuine symmetry features, but only imperfectly. After unmasking, correct and incorrect guesses were nearly balanced, and both blinding indices were close to zero. Correct guessers had descriptively higher symmetry accuracy than incorrect or uncertain guessers, but the association was small and inconclusive ($r = .18$, exact

1611 $p = .33$). Overall, the results were consistent with effective blinding, but the wide
1612 standard BBI interval prevented a conclusive determination.

1613 Participants did not reliably identify the symmetric condition. Although this is
1614 consistent with blinding, it may also limit the test of the symmetry hypothesis, because
1615 explicit attention to and awareness of the manipulated feature may be necessary
1616 for it to influence emotional experience. Symmetric coherence appeared only during
1617 the breath-hold phases, approximately half of each breathing cycle, and remained
1618 secondary to the more salient breath-coupled expansion and contraction. The null dif-
1619 ference therefore does not distinguish between an absence of an affective effect and
1620 insufficient focused exposure or awareness. We therefore plan a follow-up study that
1621 makes symmetry or particle coherence the focal stimulus dimension, manipulates it in
1622 isolation under controlled exposure, verifies that participants perceive the difference,
1623 and assesses its effects on emotional valence using physiological markers alongside
1624 direct ratings rather than relying only on retrospective altered-state ratings.

1625

1626 4.2 Limitations and future directions

1627

1628 Except for breathing-adherence PLV, the study’s outcomes relied on retrospective
1629 reports completed after each condition. The observed differences therefore establish
1630 how participants subsequently rated their experience, not whether the interven-
1631 tion produced corresponding changes in ongoing neural, autonomic, or multisensory
1632 processes.

1633 Retrospective psychometric scales remain valuable because subjective content is
1634 not directly observable. The 11D-ASC belongs to a widely used questionnaire family
1635 conceived for standardized comparison of altered-state phenomenology across induc-
1636 tion methods (Dittrich 1998; Hovmand et al. 2023). The Altered States Database
1637 demonstrates this comparative role by bringing standardized questionnaire data from
1638 pharmacological and non-pharmacological inductions into a common framework that
1639 supports direct comparisons across methods (Schmidt and Berkemeyer 2018; Prugger
1640 et al. 2022).

1641 Standardization does not remove the limitations of retrospective psychometrics.
1642 Scores remain sensitive to recall, item wording, construct coverage, and assumptions
1643 built into the instrument. Reviews of altered-state measures identify substantial varia-
1644 tion across scales and incomplete evidence for their psychometric quality and coverage,
1645 while critiques of mystical-experience instruments note their religious framing (Sanders
1646 and Zijlmans 2021; Hovmand et al. 2023; de Deus Pontual et al. 2023). The present
1647 11D-ASC results should therefore be interpreted as standardized questionnaire-defined
1648 reports that support cross-study comparison, not as a complete or modality-neutral
1649 measure of altered consciousness.

1650 Future studies should triangulate retrospective ratings with concurrent measures
1651 rather than treat any single physiological signal as a direct readout of subjective
1652 experience. Candidate approaches include EEG complexity measures associated with
1653 breathwork phenomenology (Lewis-Healey et al. 2024), the perturbational complex-
1654 ity index developed to distinguish levels of consciousness across wakefulness, sleep,
1655 anaesthesia, and disorders of consciousness (Casali et al. 2013), heart-rate entropy

1656

examined as a low-cost correlate of psychedelic states (Rosas et al. 2023), and multisensory peripersonal-space tasks that quantify changes in the represented boundary around the body (Serino 2019; Bufacchi and Iannetti 2018). These measures remain indirect and construct-specific, but convergence across subjective, neural, autonomic, and behavioural domains would provide stronger evidence than questionnaires alone.

No-report paradigms offer a separate way to reduce report-related confounds in neural measurements. They infer conscious content from previously validated behavioural or physiological signatures, but they do not eliminate the need for first-person validation (Tsuchiya et al. 2015). Laukkonen and Slagter identify neurophenomenology and no-report paradigms as complementary future directions for studying meditation states in which ordinary report may be difficult (Laukkonen and Slagter 2021).

Viscerality was designed to test whether breath-coupled changes in surrounding space can alter the felt boundary between self and environment. The spatial frame-of-reference finding provides preliminary support for this possibility, but a peripersonal-space task would offer a more direct behavioural test. Peripersonal space represents the multisensory boundary surrounding the body and can be estimated from response changes as stimuli approach or recede (Serino 2019; Bufacchi and Iannetti 2018; Karakoc et al. 2025).

A visuotactile peripersonal-space task could be administered during the 4-second breath holds without reusing the intervention’s visual features. Comparing responses after inhalation and exhalation would test whether respiratory expansion and contraction are accompanied by corresponding changes in the represented boundary around the body. A reversed-mapping condition could then test whether any shift depends on congruence between respiratory phase and spatial direction.

The particle system in the present experiment was used to convey the expansion and contraction of surrounding space in response to breathing, but particles can also encode visceral signals directly onto the body or its surroundings, or construct abstract visual configurations that convey affective states through movement, colour, shape, or synchrony. Our manipulation of dynamic particle symmetry through oscillator coupling explored only one such parameter and did not yield observable effects. Our lesson from this is that mapping aesthetic features of particle displays onto affective dimensions requires a more fine-grained approach than a single global variable assessed through altered-state questionnaires. Short aesthetic judgment tasks offer a tractable starting point, following swarm robotics research where trajectory smoothness, speed, synchronization, and spatial coherence map onto separable affective dimensions (Dietz et al. 2017; Santos and Egerstedt 2021). The immersive 3D particle environment affords considerably more degrees of freedom than planar swarm displays, with particles moving through three spatial dimensions under both local neighbour-coupling and global synchronization parameters, and testing specific feature combinations in controlled affective assessments would provide a more solid basis for these mappings than retrospective questionnaires after prolonged sessions. Because the medium is abstract rather than representational, the same particle field could in principle simultaneously encode multiple biosignals and serve as the basis for biofeedback displays that are intuitive to the user and conducive to the affective states the feedback is meant to support.

1703 The present study found that the particle system was associated with higher Pos-
1704 itive and Perceptual Effects during box breathing, alongside a smaller Distressing
1705 Effects increase. The cross-study profile provides descriptive context only and does not
1706 establish equivalence, superior tolerability, or clinical benefit. Because the system is
1707 self-administered and the session lasted 10 minutes, future studies could test whether
1708 it supports breathing practice in settings where access to a facilitator is limited or time
1709 is scarce. High-ventilation techniques are contraindicated for individuals vulnerable
1710 to panic, syncope, or hypocapnia (Fincham et al. 2023), and a slow-paced breath-
1711 coupled VR system offers a lower-intensity entry point into breathing as a therapeutic
1712 modality.

1713 One possible population for such testing is stroke survivors, given that respiratory
1714 dysfunction and sleep-disordered breathing are common after stroke and associated
1715 with poorer recovery (Patrizz et al. 2023; Seiler et al. 2019), while structured respira-
1716 tory rehabilitation remains underemphasized in clinical practice (Menezes et al. 2016).
1717 Slow-paced breathing exercises are already used to retrain respiratory function after
1718 stroke (Abdullahi et al. 2024; Yoon et al. 2022), and coupling breathing to an intuitive
1719 visual mapping of respiratory depth could engage an additional learning pathway that
1720 supports recovery of motor control over respiration. These remain illustrative possibil-
1721 ities. The present sample consisted of a homogeneous group of university psychology
1722 students, and testing usability, tolerability, and efficacy in clinical populations is a
1723 necessary next step.

1724

1725 5 Conclusion

1726

1727 During a brief self-administered box-breathing session, Viscerality was associated
1728 with closer synchronization to the instructed rhythm, a more spatially extended self-
1729 rating, and higher Positive, Perceptual, and smaller Distressing Effects than the dark-
1730 screen control. No measured outcome differed reliably between the symmetric and
1731 asymmetric VR mappings. These results come from a recruitment-naïve university
1732 sample and do not establish equivalence to longer facilitator-supported breathwork or
1733 clinical efficacy.

1734

1735 Supplementary Information

1736

1737 Online Resource 1: Supplementary methods and analyses, including deviations from
1738 preregistration, phase-locking value computation and quality screening, complete 11D-
1739 ASC results, exploratory pictograph–ASC correspondence, temporal experience tracer
1740 analyses, and reproducibility information.

1741

1742 Statements and Declarations

1743

1744 Funding: The study reported here was funded by the BIAL Foundation through Grant
1745 308/22, “Altered States of Viscerality: Augmenting Breathwork with Bio-Responsive
1746 Virtual Reality,” awarded under the 2022 Grants Programme for Scientific Research.
1747 Continued development of Viscerality received support from the Baden-Württemberg
1748

Stiftung through the Forschungswerkstatt – Deine Idee für Morgen programme for the project “Atemräume: Integration atem- und herzratenbasierter Interaktionen in therapeutische VR-Umgebungen.”	1749
Competing interests: The authors declare no competing interests.	1750
Ethics approval: The Ethics Committee of the University of Konstanz approved the study, which was conducted in accordance with the Declaration of Helsinki.	1751
Consent to participate: All participants provided informed consent before participation.	1752
Consent for publication: All participants provided written informed consent for publication and for public sharing of pseudonymized study data, as approved by the Ethics Committee of the University of Konstanz.	1753
Availability of data and materials: Pseudonymized participant-level data, the original preregistration record, statistical analysis outputs, figure-generation scripts, study materials, and reproducibility documentation are available in the versioned reproducibility package (Fejer et al. 2026b); a mirror is available at OSF: https://doi.org/10.17605/OSF.IO/64MWK .	1754
Code availability: The statistical analysis and figure-generation scripts, exact environment specifications, and replay instructions are included in the same versioned reproducibility package (Fejer et al. 2026b).	1755
Authors’ contributions: George Fejer: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. Till Holzapfel: Conceptualization, Methodology, Resources, Software, Validation, Visualization. Taru Hirvonen: Methodology, Resources, Software, Visualization. Anestis Lalidis Mateo: Methodology, Resources, Software. Johannes Blum: Methodology, Resources, Software, Validation. Michael Gaebler: Conceptualization, Project administration, Supervision, Writing – review & editing. Bigna Lenggenhager: Conceptualization, Funding acquisition, Supervision, Writing – review & editing.	1756
References	1757
Abdullahi A, Wong TWL, Ng SSM (2024) Efficacy of diaphragmatic breathing exercise on respiratory, cognitive, and motor function outcomes in patients with stroke: a systematic review and meta-analysis. <i>Frontiers in Neurology</i> 14:1233408. https://doi.org/10.3389/fneur.2023.1233408 , URL https://doi.org/10.3389/fneur.2023.1233408	1758
Abtahi P, Hough SQ, Landay JA, et al (2022) Beyond being real: A sensorimotor control perspective on interactions in virtual reality. In: <i>Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems</i> . ACM, pp 1–17, https://doi.org/10.1145/3491102.3517706 , URL https://doi.org/10.1145/3491102.3517706	1759
Acebrón JA, Bonilla LL, Pérez Vicente CJ, et al (2005) The kuramoto model: A simple paradigm for synchronization phenomena. <i>Reviews of modern physics</i> 77(1):137–185. https://doi.org/10.1103/RevModPhys.77.137	1760

1795 Adler D, Herbelin B, Similowski T, et al (2014) Breathing and sense of self:
1796 Visuo-respiratory conflicts alter body self-consciousness. *Respiratory Physiology*
1797 & *Neurobiology* 203:68–74. <https://doi.org/10.1016/j.resp.2014.08.003>, URL <https://doi.org/10.1016/j.resp.2014.08.003>
1798
1799
1800 Ahmed A, Devi RG, Priya AJ (2021) Effect of box breathing technique on lung
1801 function test. *Journal of Pharmaceutical Research International* 33(58A):25–31.
1802 <https://doi.org/10.9734/JPRI/2021/v33i58A34085>, URL <https://doi.org/10.9734/JPRI/2021/v33i58A34085>
1803
1804
1805 Alhammad SA (2024) Advocating for action: Exploring the potential of virtual reality
1806 in breathing exercise – a review of the clinical applications. *Patient Preference and*
1807 *Adherence* 18:695–707. <https://doi.org/10.2147/ppa.s451609>, URL <https://doi.org/10.2147/ppa.s451609>
1808
1809
1810 Antle AN, Corness G, Droumeva M (2009) What the body knows: Exploring the
1811 benefits of embodied metaphors in hybrid physical digital environments. *Interacting*
1812 *with Computers* 21(1–2):66–75. <https://doi.org/10.1016/j.intcom.2008.10.005>, URL
1813 <https://doi.org/10.1016/j.intcom.2008.10.005>
1814
1815 Aronoff J (2006) How we recognize angry and happy emotion in people,
1816 places, and things. *Cross-Cultural Research* 40(1):83–105. <https://doi.org/10.1177/1069397105282597>, URL <https://doi.org/10.1177/1069397105282597>
1817
1818 Ashhad S, Kam K, Del Negro CA, et al (2022) Breathing rhythm and pattern and
1819 their influence on emotion. *Annual Review of Neuroscience* 45:223–247. <https://doi.org/10.1146/annurev-neuro-090121-014424>
1820
1821
1822 Aspell JE, Heydrich L, Marillier G, et al (2013) Turning body and self inside out. *Psy-*
1823 *chological Science* 24(12):2445–2453. <https://doi.org/10.1177/0956797613498395>,
1824 URL <https://doi.org/10.1177/0956797613498395>
1825
1826 Azzalini D, Rebollo I, Tallon-Baudry C (2019) Visceral signals shape brain dynamics
1827 and cognition. *Trends in Cognitive Sciences* 23(6):488–509. <https://doi.org/10.1016/j.tics.2019.03.007>
1828
1829
1830 Bahi C, Irrmischer M, Franken K, et al (2024) Effects of conscious connected breathing
1831 on cortical brain activity, mood and state of consciousness in healthy adults. *Current*
1832 *Psychology* 43(12):10578–10589. <https://doi.org/10.1007/s12144-023-05119-6>
1833
1834 Balban MY, Neri E, Kogon MM, et al (2023) Brief structured respiration practices
1835 enhance mood and reduce physiological arousal. *Cell Reports Medicine* 4(1):100895.
1836 <https://doi.org/10.1016/j.xcrm.2022.100895>
1837
1838 Bang H, Ni L, Davis CE (2004) Assessment of blinding in clinical trials. *Controlled*
1839 *Clinical Trials* 25(2):143–156. <https://doi.org/10.1016/j.cct.2003.10.016>
1840

- Banushi B, Brendle M, Ragnhildstveit A, et al (2023) Breathwork interventions for adults with clinically diagnosed anxiety disorders: A scoping review. *Brain Sciences* 13(2):256. <https://doi.org/10.3390/brainsci13020256>
- Barton AC, Do M, Sheen J, et al (2024) The restorative and state enhancing potential of abstract fractal-like imagery and interactive mindfulness interventions in virtual reality. *Virtual Reality* 28(1):53. <https://doi.org/10.1007/s10055-023-00916-7>
- Bartram L, Nakatani A (2009) Distinctive parameters of expressive motion. In: *Computational Aesthetics in Graphics, Visualization, and Imaging*. The Eurographics Association, pp 129–136, <https://doi.org/10.2312/COMPAESTH/COMPAESTH09/129-136>, URL <http://diglib.eg.org/handle/10.2312/COMPAESTH.COMPAESTH09.129-136>
- Belli F, Felisatti A, Fischer MH (2021) BreaThink: breathing affects production and perception of quantities. *Experimental Brain Research* 239(8):2489–2499. <https://doi.org/10.1007/s00221-021-06147-z>, URL <https://doi.org/10.1007/s00221-021-06147-z>
- Bentley TGK, D’Andrea-Penna G, Rakic M, et al (2023) Breathing practices for stress and anxiety reduction: Conceptual framework of implementation guidelines based on a systematic review of the published literature. *Brain Sciences* 13(12):1612. <https://doi.org/10.3390/brainsci13121612>
- Bertamini M, Makin ADJ, Rampone G (2013) Implicit association of symmetry with positive valence, high arousal and simplicity. *i-Perception* 4(5):317–327. <https://doi.org/10.1068/i0601jw>
- Betka S, Canzoneri E, Adler D, et al (2020) Mechanisms of the breathing contribution to bodily self-consciousness in healthy humans: Lessons from machine-assisted breathing? *Psychophysiology* 57(8):e13564. <https://doi.org/10.1111/psyp.13564>, URL <https://doi.org/10.1111/psyp.13564>
- Billock VA, Tsou BH (2012) Elementary visual hallucinations and their relationships to neural pattern-forming mechanisms. *Psychological Bulletin* 138(4):744–774. <https://doi.org/10.1037/a0027580>, URL <https://doi.org/10.1037/a0027580>
- Blazhenkova O, Kumar MM (2018) Angular versus curved shapes: Correspondences and emotional processing. *Perception* 47(1):67–89. <https://doi.org/10.1177/0301006617731048>, URL <https://doi.org/10.1177/0301006617731048>
- Blum J, Rockstroh C, Göritz AS (2019) Heart rate variability biofeedback based on slow-paced breathing with immersive virtual reality nature scenery. *Frontiers in psychology* 10:2172. <https://doi.org/10.3389/fpsyg.2019.02172>
- Blum J, Rockstroh C, Göritz AS (2020) Development and pilot test of a virtual reality respiratory biofeedback approach. *Applied Psychophysiology and Biofeedback*

1887 45(3):153–163. <https://doi.org/10.1007/s10484-020-09468-x>
1888
1889 Boiten FA, Frijda NH, Wientjes CJE (1994) Emotions and respiratory patterns: review
1890 and critical analysis. *International Journal of Psychophysiology* 17(2):103–128.
1891 [https://doi.org/10.1016/0167-8760\(94\)90027-2](https://doi.org/10.1016/0167-8760(94)90027-2)
1892
1893 Bossenbroek R, Wols A, Weerdmeester J, et al (2020) Efficacy of a virtual reality
1894 biofeedback game (DEEP) to reduce anxiety and disruptive classroom behavior:
1895 single-case study. *JMIR Mental Health* 7(3):e16066. <https://doi.org/10.2196/16066>,
1896 URL <https://doi.org/10.2196/16066>
1897
1898 Bufacchi RJ, Iannetti GD (2018) An action field theory of peripersonal space. *Trends*
1899 *in Cognitive Sciences* 22(12):1076–1090. <https://doi.org/10.1016/j.tics.2018.09.004>
1900
1901 Burger B, Toiviainen P (2020) See how it feels to move: Relationships between
1902 movement characteristics and perception of emotions in dance. *Human Technol-*
1903 *ogy* 16(3):233–256. <https://doi.org/10.17011/ht/urn.202011256764>, URL [https://](https://doi.org/10.17011/ht/urn.202011256764)
1904 doi.org/10.17011/ht/urn.202011256764
1905
1906 Casali AG, Gosseries O, Rosanova M, et al (2013) A theoretically based index of
1907 consciousness independent of sensory processing and behavior. *Science Translational*
1908 *Medicine* 5(198):198ra105. <https://doi.org/10.1126/scitranslmed.3006294>
1909
1910 Chan PYS, Lee LY, Davenport PW (2024) Neural mechanisms of respiratory intero-
1911 ception. *Autonomic Neuroscience: Basic and Clinical* 253:103181. [https://doi.org/](https://doi.org/10.1016/j.autneu.2024.103181)
1912 [10.1016/j.autneu.2024.103181](https://doi.org/10.1016/j.autneu.2024.103181)
1913
1914 Chang E, Kim HT, Yoo B (2020) Virtual reality sickness: a review of causes and
1915 measurements. *International Journal of Human–Computer Interaction* 36(17):1658–
1916 1682. <https://doi.org/10.1080/10447318.2020.1778351>
1917
1918 Cheiran JFP, Bandeira DR, Pimenta MS (2025) Measuring the key components of the
1919 user experience in immersive virtual reality environments. *Frontiers in Virtual Real-*
1920 *ity* 6:1585614. <https://doi.org/10.3389/frvir.2025.1585614>, URL [https://doi.org/10.](https://doi.org/10.3389/frvir.2025.1585614)
1921 [3389/frvir.2025.1585614](https://doi.org/10.3389/frvir.2025.1585614)
1922
1923 Cortez-Vázquez G, Adriaanse M, Burchell GL, et al (2024) Virtual reality breath-
1924 ing interventions for mental health: A systematic review and meta-analysis of
1925 randomized controlled trials. *Applied Psychophysiology and Biofeedback* 49(1):1–
1926 21. <https://doi.org/10.1007/s10484-023-09611-4>, URL [https://doi.org/10.1007/](https://doi.org/10.1007/s10484-023-09611-4)
1927 [s10484-023-09611-4](https://doi.org/10.1007/s10484-023-09611-4)
1928
1929 Dambrun M (2016) When the dissolution of perceived body boundaries elicits happi-
1930 ness: The effect of selflessness induced by a body scan meditation. *Consciousness*
1931 *and Cognition* 46:89–98. <https://doi.org/10.1016/j.concog.2016.09.013>
1932

Dambrun M (2023) Beyond the body/self-consciousness matrix: Scale of the physical frame of reference and self-transcendence experience. <i>Psychology of Consciousness: Theory, Research, and Practice</i> 10(3):292–312. https://doi.org/10.1037/cns0000314 , URL https://doi.org/10.1037/cns0000314	1933 1934 1935 1936 1937
Damiano C, Gayen P, Rezanejad M, et al (2023) Anger is red, sadness is blue: Emotion depictions in abstract visual art by artists and non-artists. <i>Journal of Vision</i> 23(4):1. https://doi.org/10.1167/jov.23.4.1 , URL https://doi.org/10.1167/jov.23.4.1	1938 1939 1940 1941
de Deus Pontual AA, Senhorini HG, Corradi-Webster CM, et al (2023) Systematic review of psychometric instruments used in research with psychedelics. <i>Journal of Psychoactive Drugs</i> 55(3):359–368. https://doi.org/10.1080/02791072.2022.2079108	1942 1943 1944 1945
Dietz G, Jane L. E., Washington P, et al (2017) Human perception of swarm robot motion. In: <i>Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems</i> . ACM, pp 2520–2527, https://doi.org/10.1145/3027063.3053220 , URL https://doi.org/10.1145/3027063.3053220	1946 1947 1948 1949
Dittrich A (1998) The standardized psychometric assessment of altered states of consciousness (ascs) in humans. <i>Pharmacopsychiatry</i> 31(S2):80–84. https://doi.org/10.1055/s-2007-979351	1950 1951 1952 1953
van Doorn JB, van den Bergh D, Bohm U, et al (2021) The JASP guidelines for conducting and reporting a bayesian analysis. <i>Psychonomic Bulletin & Review</i> 28(3):813–826. https://doi.org/10.3758/s13423-020-01798-5	1954 1955 1956 1957
Ekman P (1993) Facial expression and emotion. <i>American psychologist</i> 48(4):384–392. https://doi.org/10.1037/0003-066X.48.4.384	1958 1959 1960
Engelen T, Solcà M, Tallon-Baudry C (2023) Interoceptive rhythms in the brain. <i>Nature Neuroscience</i> 26(10):1670–1684. https://doi.org/10.1038/s41593-023-01425-1	1961 1962 1963 1964
Fejer G, Holzapfel T, Gómez-Emilsson A, et al (2025) Viscerality: A bio-responsive vr system for breath-based interactions and coupled oscillator dynamics to augment altered states of consciousness. In: <i>Mensch und Computer 2025 Workshopband, Gesellschaft für Informatik e.V.</i> , https://doi.org/10.18420/muc2025-mci-ws11-174 , URL https://dl.gi.de/handle/20.500.12116/46829	1965 1966 1967 1968 1969
Fejer G, Holzapfel T, Hirvonen T, et al (2026a) Breathing space: Spatial mapping of breath and cardiac biofeedback for affective state representation and coherence training in viscerality. In: <i>Proceedings of the 1st International Conference on Human-Computer Interaction in the Alps</i> . ACM, pp 79–86, https://doi.org/10.1145/3780045.3780061	1970 1971 1972 1973 1974 1975
Fejer G, Holzapfel T, Hirvonen T, et al (2026b) Viscerality: Bioresponsive virtual reality for box breathing and altered-state experience: Study 3 reproducibility	1976 1977 1978

1979 package. <https://doi.org/10.5281/zenodo.22073981>, URL <https://doi.org/10.5281/zenodo.22073981>

1980

1981

1982 Fincham GW, Strauss C, Montero-Marin J, et al (2023) Effect of breathwork on

1983 stress and mental health: A meta-analysis of randomised-controlled trials. *Scientific*

1984 *Reports* 13(1):432. <https://doi.org/10.1038/s41598-022-27247-y>

1985

1986 Fincham GW, Epel E, Colasanti A, et al (2024) Effects of brief remote high ven-

1987 tilation breathwork with retention on mental health and wellbeing: a randomised

1988 placebo-controlled trial. *Scientific Reports* 14(1):16893. <https://doi.org/10.1038/s41598-024-64254-7>

1989

1990 Fincham GW, Caddy E, Kartar AA, et al (2026) An exploratory study of breathwork-

1991 induced altered states of consciousness in experienced practitioners: the airways

1992 to alteration (A2A) trial. *Frontiers in Psychology* 17:1851882. <https://doi.org/10.3389/fpsyg.2026.1851882>

1993

1994

1995 Gerritsen RJS, Band GPH (2018) Breath of life: The respiratory vagal stimulation

1996 model of contemplative activity. *Frontiers in Human Neuroscience* 12:397. <https://doi.org/10.3389/fnhum.2018.00397>

1997

1998

1999 Glowacki DR (2024) VR models of death and psychedelics: An aesthetic paradigm

2000 for design beyond day-to-day phenomenology. *Frontiers in Virtual Reality*

2001 4:1286950. <https://doi.org/10.3389/frvir.2023.1286950>, URL <https://doi.org/10.3389/frvir.2023.1286950>

2002

2003

2004 Glowacki DR, Wonnacott MD, Freire R, et al (2020) Isness: Using multi-person VR

2005 to design peak mystical type experiences comparable to psychedelics. In: *Proceed-*

2006 *ings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM,

2007 pp 1–14, <https://doi.org/10.1145/3313831.3376649>, URL <https://doi.org/10.1145/3313831.3376649>

2008

2009

2010 Glowacki DR, Williams RR, Wonnacott MD, et al (2022) Group VR experiences can

2011 produce ego attenuation and connectedness comparable to psychedelics. *Scientific*

2012 *Reports* 12(1):8995. <https://doi.org/10.1038/s41598-022-12637-z>, URL <https://doi.org/10.1038/s41598-022-12637-z>

2013

2014

2015 Goral O, Wald IY, Maimon A, et al (2024) Enhancing interoceptive sensibility

2016 through exteroceptive-interoceptive sensory substitution. *Scientific Reports*

2017 14(1):14855. <https://doi.org/10.1038/s41598-024-63231-4>, URL <https://doi.org/10.1038/s41598-024-63231-4>

2018

2019

2020 Guy M, Normand JM, Jeunet-Kelway C, et al (2023) The sense of embodiment

2021 in virtual reality and its assessment methods. *Frontiers in Virtual Reality*

2022 4:1141683. <https://doi.org/10.3389/frvir.2023.1141683>, URL <https://doi.org/10.3389/frvir.2023.1141683>

2023

2024

Hanley AW, Garland EL (2019) Spatial frame of reference as a phenomenological feature of self-transcendence: Measurement and manipulation through mindfulness meditation. *Psychology of Consciousness: Theory, Research, and Practice* 6(4):329–345. <https://doi.org/10.1037/cns0000204>

Hanley AW, Dambrun M, Garland EL (2020) Effects of mindfulness meditation on self-transcendent states: Perceived body boundaries and spatial frames of reference. *Mindfulness* 11(5):1194–1203. <https://doi.org/10.1007/s12671-020-01330-9>

Havenith MN, Leidenberger M, Brasanac J, et al (2025) Decreased co2 saturation during circular breathwork supports emergence of altered states of consciousness. *Communications Psychology* 3(1):59. <https://doi.org/10.1038/s44271-025-00247-0>

Homma I, Masaoka Y (2008) Breathing rhythms and emotions. *Experimental Physiology* 93(9):1011–1021. <https://doi.org/10.1113/expphysiol.2008.042424>

Hovmand OR, Poulsen ED, Arnfred S (2023) Assessment of the acute subjective psychedelic experience: A review of patient-reported outcome measures in clinical research on classical psychedelics. *Journal of Psychopharmacology* 38(1):19–32. <https://doi.org/10.1177/02698811231200019>

Hovmand OR, Madsen M, Fisher PM, et al (2024) Altered states of consciousness in danish healthy volunteers and recreational users of psilocybin and the possible impact of setting and intention: Danish validation of the five-dimensional altered states of consciousness questionnaire. *Journal of Psychopharmacology* 38(9):924–932. <https://doi.org/10.1177/02698811241269669>

Hruby H, Schmidt S, Feinstein JS, et al (2024) Induction of altered states of consciousness during Floatation-REST is associated with the dissolution of body boundaries and the distortion of subjective time. *Scientific Reports* 14(1):9316. <https://doi.org/10.1038/s41598-024-59642-y>, URL <https://doi.org/10.1038/s41598-024-59642-y>

Jachs B (2021) The neurophenomenology of meditative states: Introducing temporal experience tracing to capture subjective experience states and their neural correlates. PhD thesis, University of Cambridge, <https://doi.org/10.17863/CAM.80153>, URL <https://www.repository.cam.ac.uk/handle/1810/332709>

Jerath R, Beveridge C (2020) Respiratory rhythm, autonomic modulation, and the spectrum of emotions: The future of emotion recognition and modulation. *Frontiers in Psychology* 11:1980. <https://doi.org/10.3389/fpsyg.2020.01980>

Johnson ME (2023) Qualia formalism and a symmetry theory of valence. *OpenTheory*, URL https://opentheory.net/Qualia_Formalism_and_a_Symmetry_Theory_of_Valence.pdf

Kaduk J, Cavdan M, Drewing K, et al (2024) From one to many: How active robot swarm sizes influence human cognitive processes. In: 2024 33rd IEEE International

2071 Conference on Robot and Human Interactive Communication (RO-MAN). IEEE,
 2072 pp 1207–1212, <https://doi.org/10.1109/RO-MAN60168.2024.10731232>, URL <https://doi.org/10.1109/RO-MAN60168.2024.10731232>
 2073
 2074
 2075 Karakoc C, Lucifora C, Massimino S, et al (2025) Extending peri-personal
 2076 space in immersive virtual reality: A systematic review. *Virtual Worlds*
 2077 4(1):5. <https://doi.org/10.3390/virtualworlds4010005>, URL <https://doi.org/10.3390/virtualworlds4010005>
 2078
 2079 Kartar AA, Horinouchi T, Örsik B, et al (2025) Neurobiological substrates of altered
 2080 states of consciousness induced by high ventilation breathwork accompanied by
 2081 music. *PLOS One* 20(8):e0329411. <https://doi.org/10.1371/journal.pone.0329411>,
 2082 URL <https://doi.org/10.1371/journal.pone.0329411>
 2083
 2084 Kaufeld M, Bourdeinik J, Prinz LM, et al (2022) Emotions are associated with the
 2085 genesis of visually induced motion sickness in virtual reality. *Experimental Brain*
 2086 *Research* 240(10):2757–2771. <https://doi.org/10.1007/s00221-022-06454-z>
 2087
 2088 Khatin-Zadeh O, Eskandari Z, Vahdat S, et al (2019) Why are motions effective in
 2089 describing emotions? *Polish Psychological Bulletin* 50(2):119–124. <https://doi.org/10.24425/ppb.2019.126027>
 2090
 2091 Klüver H (1966) *Mescal, and Mechanisms of Hallucinations*. University of Chicago
 2092 Press, Chicago, reprint edition; originally published 1928
 2093
 2094 Kodithuwakku Arachchige SNK, Chander H, Freeman H, et al (2022) The walls are
 2095 closing in: postural responses to a virtual reality claustrophobic simulation. *Clinical*
 2096 *and Translational Neuroscience* 6(2):15. <https://doi.org/10.3390/ctn6020015>
 2097
 2098 Kolahi J, Bang H, Park J (2009) Towards a proposal for assessment of blinding success
 2099 in clinical trials: Up-to-date review. *Community Dentistry and Oral Epidemiology*
 2100 37(6):477–484. <https://doi.org/10.1111/j.1600-0528.2009.00494.x>
 2101
 2102 Kreibig SD (2010) Autonomic nervous system activity in emotion: A review. *Biological*
 2103 *Psychology* 84(3):394–421. <https://doi.org/10.1016/j.biopsycho.2010.03.010>
 2104
 2105 Kruschke JK (2021) Bayesian analysis reporting guidelines. *Nature Human Behaviour*
 2106 5(10):1282–1291. <https://doi.org/10.1038/s41562-021-01177-7>
 2107
 2108 Kytikova O, Novgorodtseva T, Denisenko Y, et al (2021) Thermosensory transient
 2109 receptor potential ion channels and asthma. *Biomedicines* 9(7):816. <https://doi.org/10.3390/biomedicines9070816>
 2110
 2111 Laukkonen RE, Slagter HA (2021) From many to (n)one: Meditation and the plasticity
 2112 of the predictive mind. *Neuroscience & Biobehavioral Reviews* 128:199–217. <https://doi.org/10.1016/j.neubiorev.2021.06.021>
 2113
 2114
 2115
 2116

Lawrence DW, Carhart-Harris R, Griffiths R, et al (2022) Phenomenology and content of the inhaled N, N-dimethyltryptamine (N, N-DMT) experience. <i>Scientific Reports</i> 12(1):8562. https://doi.org/10.1038/s41598-022-11999-8 , URL https://doi.org/10.1038/s41598-022-11999-8	2117 2118 2119 2120 2121
Lenggenhager B, Tadi T, Metzinger T, et al (2007) Video ergo sum: Manipulating bodily self-consciousness. <i>Science</i> 317(5841):1096–1099. https://doi.org/10.1126/science.1143439 , URL https://doi.org/10.1126/science.1143439	2122 2123 2124 2125
Lewis-Healey E, Tagliazucchi E, Canales-Johnson A, et al (2024) Breathwork-induced psychedelic experiences modulate neural dynamics. <i>Cerebral Cortex</i> 34(8):bhae347. https://doi.org/10.1093/cercor/bhae347	2126 2127 2128 2129
Makin ADJ, Wilton MM, Pecchinenda A, et al (2012) Symmetry perception and affective responses: A combined eeg/emg study. <i>Neuropsychologia</i> 50(14):3250–3261. https://doi.org/10.1016/j.neuropsychologia.2012.10.003	2130 2131 2132 2133
Maravita A, Iriki A (2004) Tools for the body (schema). <i>Trends in Cognitive Sciences</i> 8(2):79–86. https://doi.org/10.1016/j.tics.2003.12.008	2134 2135 2136
Marcolin F, Wally Scurati G, Ulrich L, et al (2021) Affective virtual reality: How to design artificial experiences impacting human emotions. <i>IEEE Computer Graphics and Applications</i> 41(6):171–178. https://doi.org/10.1109/MCG.2021.3115015 , URL https://doi.org/10.1109/MCG.2021.3115015	2137 2138 2139 2140
McGraw KO, Wong SP (1992) A common language effect size statistic. <i>Psychological Bulletin</i> 111(2):361–365. https://doi.org/10.1037/0033-2909.111.2.361 , URL https://doi.org/10.1037/0033-2909.111.2.361	2141 2142 2143 2144
Menezes KKP, Nascimento LR, Ada L, et al (2016) Respiratory muscle training increases respiratory muscle strength and reduces respiratory complications after stroke: a systematic review. <i>Journal of Physiotherapy</i> 62(3):138–144. https://doi.org/10.1016/j.jphys.2016.05.014 , URL https://doi.org/10.1016/j.jphys.2016.05.014	2145 2146 2147 2148 2149
Millière R (2020) The varieties of selflessness. <i>Philosophy and the Mind Sciences</i> 1(I):1–41. https://doi.org/10.33735/phimisci.2020.i.48 , URL https://doi.org/10.33735/phimisci.2020.i.48	2150 2151 2152 2153
Mogan R, Fischer R, Bulbulia JA (2017) To be in synchrony or not? a meta-analysis of synchrony’s effects on behavior, perception, cognition and affect. <i>Journal of Experimental Social Psychology</i> 72:13–20. https://doi.org/10.1016/j.jesp.2017.03.009	2154 2155 2156 2157
Monti A, Porciello G, Tieri G, et al (2020) The “embreathment” illusion highlights the role of breathing in corporeal awareness. <i>Journal of Neurophysiology</i> 123(1):420–427. https://doi.org/10.1152/jn.00617.2019 , URL https://doi.org/10.1152/jn.00617.2019	2158 2159 2160 2161 2162

2163 Morey RD, Rouder JN (2024) BayesFactor: Computation of Bayes Factors for Com-
 2164 mon Designs. <https://doi.org/10.32614/CRAN.package.BayesFactor>, URL <https://CRAN.R-project.org/package=BayesFactor>, r package version 0.9.12-4.7
 2165
 2166
 2167 Mori K, Sakano H (2022a) Neural circuitry for stress information of environmental
 2168 and internal odor worlds. *Frontiers in Behavioral Neuroscience* 16:943647. <https://doi.org/10.3389/fnbeh.2022.943647>
 2169
 2170
 2171 Mori K, Sakano H (2022b) Processing of odor information during the respiratory cycle
 2172 in mice. *Frontiers in Neural Circuits* 16:861800. [https://doi.org/10.3389/fncir.2022.](https://doi.org/10.3389/fncir.2022.861800)
 2173 [861800](https://doi.org/10.3389/fncir.2022.861800)
 2174
 2175 Muth C, Carbon CC (2013) The aesthetic aha: On the pleasure of having insights into
 2176 gestalt. *Acta psychologica* 144(1):25–30. [https://doi.org/10.1016/j.actpsy.2013.05.](https://doi.org/10.1016/j.actpsy.2013.05.001)
 2177 [001](https://doi.org/10.1016/j.actpsy.2013.05.001)
 2178
 2179 Pancini E, Di Natale AF, Villani D (2025) Breathing in virtual reality for promoting
 2180 mental health: a scoping review. *Virtual Reality* 29(1):29. [https://doi.org/10.1007/](https://doi.org/10.1007/s10055-024-01096-8)
 2181 [s10055-024-01096-8](https://doi.org/10.1007/s10055-024-01096-8), URL <https://doi.org/10.1007/s10055-024-01096-8>
 2182
 2183 Patrizz A, El Hamamy A, Maniskas M, et al (2023) Stroke-induced respira-
 2184 tory dysfunction is associated with cognitive decline. *Stroke* 54(7):1863–1874.
 2185 <https://doi.org/10.1161/STROKEAHA.122.041239>, URL [https://doi.org/10.1161/](https://doi.org/10.1161/STROKEAHA.122.041239)
 2186 [STROKEAHA.122.041239](https://doi.org/10.1161/STROKEAHA.122.041239)
 2187
 2188 Pecchinenda A, Bertamini M, Makin AD, et al (2014) The pleasantness of visual
 2189 symmetry: always, never or sometimes. *PLoS ONE* 9(3):e92685. [https://doi.org/10.](https://doi.org/10.1371/journal.pone.0092685)
 2190 [1371/journal.pone.0092685](https://doi.org/10.1371/journal.pone.0092685)
 2191
 2192 Petkova VI, Ehrsson HH (2008) If I were you: Perceptual illusion of body swapping.
 2193 *PLoS ONE* 3(12):e3832. <https://doi.org/10.1371/journal.pone.0003832>, URL <https://doi.org/10.1371/journal.pone.0003832>
 2194
 2195 Petrizzo I, Mikellidou K, Avraam S, et al (2024) Reshaping the periper-
 2196 sonal space in virtual reality. *Scientific Reports* 14(1). [https://doi.org/10.1038/](https://doi.org/10.1038/s41598-024-52383-y)
 2197 [s41598-024-52383-y](https://doi.org/10.1038/s41598-024-52383-y), URL <https://doi.org/10.1038/s41598-024-52383-y>
 2198
 2199 Pöhlmann KMT, Föcker J, Dickinson P, et al (2021) The effect of motion direction
 2200 and eccentricity on vection, VR sickness and head movements in virtual reality.
 2201 *Multisensory Research* 34(6):623–662. <https://doi.org/10.1163/22134808-bja10049>
 2202
 2203 Pollick FE, Paterson HM, Bruderlin A, et al (2001) Perceiving affect from arm move-
 2204 ment. *Cognition* 82(2):B51–B61. [https://doi.org/10.1016/S0010-0277\(01\)00147-0](https://doi.org/10.1016/S0010-0277(01)00147-0),
 2205 URL [https://doi.org/10.1016/S0010-0277\(01\)00147-0](https://doi.org/10.1016/S0010-0277(01)00147-0)
 2206
 2207 Prpa M, Fdili-Alaoui S, Schiphorst T, et al (2020) Articulating experience. In: Pro-
 2208 ceedings of the 2020 CHI Conference on Human Factors in Computing Systems.

ACM, pp 1–14, https://doi.org/10.1145/3313831.3376664 , URL https://doi.org/10.1145/3313831.3376664	2209 2210 2211
Prugger J, Derdiyok E, Dinkelacker J, et al (2022) The altered states database: Psychometric data from a systematic literature review. <i>Scientific Data</i> 9(1):720. https://doi.org/10.1038/s41597-022-01822-4	2212 2213 2214 2215
Qin Z, Yu Y, Gu H, et al (2024) Sas macro programme for Bang’s Blinding Index to assess and visualise the success of blinding in randomised controlled trials. <i>General Psychiatry</i> 37:e101578. https://doi.org/10.1136/gpsych-2024-101578	2216 2217 2218 2219
Ramachandran VS, Hirstein W (1999) The science of art: A neurological theory of aesthetic experience. <i>Journal of Consciousness Studies</i> 6(6–7):15–31	2220 2221 2222
Reeves WT (1983) Particle systems—a technique for modeling a class of fuzzy objects. <i>ACM Transactions on Graphics</i> 2(2):91–108. https://doi.org/10.1145/357318.357320 , URL https://doi.org/10.1145/357318.357320	2223 2224 2225 2226
Reynolds CW (1987) Flocks, herds, and schools: A distributed behavioral model. In: <i>Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques</i> . ACM, pp 25–34, https://doi.org/10.1145/37401.37406 , URL https://doi.org/10.1145/37401.37406	2227 2228 2229 2230
Rockstroh C, Blum J, Göritz AS (2019) Virtual reality in the application of heart rate variability biofeedback. <i>International Journal of Human-Computer Studies</i> 130:209–220. https://doi.org/10.1016/j.ijhcs.2019.06.011 , URL https://doi.org/10.1016/j.ijhcs.2019.06.011	2231 2232 2233 2234 2235
Rockstroh C, Blum J, Göritz AS (2021) A mobile VR-based respiratory biofeedback game to foster diaphragmatic breathing. <i>Virtual Reality</i> 25(2):539–552. https://doi.org/10.1007/s10055-020-00471-5 , URL https://doi.org/10.1007/s10055-020-00471-5	2236 2237 2238 2239 2240
Rodrigues FA, Peron TKDM, Ji P, et al (2016) The kuramoto model in complex networks. <i>Physics Reports</i> 610:1–98. https://doi.org/10.1016/j.physrep.2015.10.008	2241 2242 2243
Rogers B, Graham M (1979) Motion parallax as an independent cue for depth perception. <i>Perception</i> 8(2):125–134. https://doi.org/10.1068/p080125 , URL https://doi.org/10.1068/p080125	2244 2245 2246 2247
Romeo B, Kervadec E, Fauvel B, et al (2025) The intensity of the psychedelic experience is reliably associated with clinical improvements: A systematic review and meta-analysis. <i>Neuroscience & Biobehavioral Reviews</i> 172:106086. https://doi.org/10.1016/j.neubiorev.2025.106086 , URL https://doi.org/10.1016/j.neubiorev.2025.106086	2248 2249 2250 2251 2252 2253 2254

2255 de Rooij A, Broekens J, Lamers MH (2013) Abstract expressions of affect. Inter-
 2256 national Journal of Synthetic Emotions 4(1):1–31. [https://doi.org/10.4018/jse.](https://doi.org/10.4018/jse.2013010101)
 2257 [2013010101](https://doi.org/10.4018/jse.2013010101), URL <https://doi.org/10.4018/jse.2013010101>
 2258
 2259 van Rooij M, Lobel A, Harris O, et al (2016) DEEP: A biofeedback virtual reality game
 2260 for children at-risk for anxiety. In: Proceedings of the 2016 CHI Conference Extended
 2261 Abstracts on Human Factors in Computing Systems. ACM, pp 1989–1997, [https://](https://doi.org/10.1145/2851581.2892452)
 2262 doi.org/10.1145/2851581.2892452, URL [https://doi.](https://doi.org/10.1145/2851581.2892452)
 2263 [org/10.1145/2851581.2892452](https://doi.org/10.1145/2851581.2892452)
 2264 Rook C, Panda S, Sice P, et al (2026) Heart garden: Visualizing heart rate data using
 2265 the metaphor of a garden. International Journal of Human–Computer Interaction
 2266 42(1):114–130. <https://doi.org/10.1080/10447318.2025.2504201>, URL [https://doi.](https://doi.org/10.1080/10447318.2025.2504201)
 2267 [org/10.1080/10447318.2025.2504201](https://doi.org/10.1080/10447318.2025.2504201)
 2268
 2269 Rosas FE, Mediano PAM, Timmermann C, et al (2023) The entropic heart: Track-
 2270 ing the psychedelic state via heart rate dynamics. bioRxiv p 2023.11.07.566008.
 2271 <https://doi.org/10.1101/2023.11.07.566008>, URL [https://doi.org/10.1101/2023.11.](https://doi.org/10.1101/2023.11.07.566008)
 2272 [07.566008](https://doi.org/10.1101/2023.11.07.566008)
 2273
 2274 Roseman L, Nutt DJ, Carhart-Harris RL (2018) Quality of acute psychedelic experi-
 2275 ence predicts therapeutic efficacy of psilocybin for treatment-resistant depression.
 2276 Frontiers in Pharmacology 8:974. <https://doi.org/10.3389/fphar.2017.00974>
 2277
 2278 Rubin DS, Morgan RC, Pinchbeck D (2010) Psychedelic: Optical and Visionary Art
 2279 Since the 1960s. MIT Press, <https://doi.org/10.5860/choice.48-0074>
 2280
 2281 Sabnis AS, Shadid M, Yost GS, et al (2008) Human lung epithelial cells express a
 2282 functional cold-sensing TRPM8 variant. American Journal of Respiratory Cell and
 2283 Molecular Biology 39(4):466–474. <https://doi.org/10.1165/rcmb.2007-0440OC>
 2284
 2285 Sanders JW, Zijlmans J (2021) Moving past mysticism in psychedelic science. ACS
 2286 Pharmacology & Translational Science 4(3):1253–1255. [https://doi.org/10.1021/](https://doi.org/10.1021/acsptsci.1c00097)
 2287 [acsptsci.1c00097](https://doi.org/10.1021/acsptsci.1c00097), URL <https://doi.org/10.1021/acsptsci.1c00097>
 2288
 2289 Santos M, Egerstedt M (2021) From motions to emotions: Can the fundamental emo-
 2290 tions be expressed in a robot swarm? International Journal of Social Robotics
 2291 13(4):751–764. <https://doi.org/10.1007/s12369-020-00665-6>, URL [https://doi.org/](https://doi.org/10.1007/s12369-020-00665-6)
 2292 [10.1007/s12369-020-00665-6](https://doi.org/10.1007/s12369-020-00665-6)
 2293
 2294 Sarkar AA (2017) Functional correlation between breathing and emotional states.
 2295 MOJ Anatomy & Physiology 3(5):157–158. [https://doi.org/10.15406/mojap.2017.](https://doi.org/10.15406/mojap.2017.03.00108)
 2296 [03.00108](https://doi.org/10.15406/mojap.2017.03.00108)
 2297
 2298 Schmidt TT, Berkemeyer H (2018) The altered states database: Psychometric data of
 2299 altered states of consciousness. Frontiers in Psychology 9:1028. [https://doi.org/10.](https://doi.org/10.3389/fpsyg.2018.01028)
 2300 [3389/fpsyg.2018.01028](https://doi.org/10.3389/fpsyg.2018.01028)

Seiler A, Camilo M, Korostovtseva L, et al (2019) Prevalence of sleep-disordered breathing after stroke and TIA. *Neurology* 92(7):e648–e654. <https://doi.org/10.1212/WNL.0000000000006904>, URL <https://doi.org/10.1212/WNL.0000000000006904>

Serino A (2019) Peripersonal space (PPS) as a multisensory interface between the individual and the environment, defining the space of the self. *Neuroscience & Biobehavioral Reviews* 99:138–159. <https://doi.org/10.1016/j.neubiorev.2019.01.016>

Shamekhi A, Bickmore T (2018) Breathe deep. In: *Proceedings of the 12th EAI International Conference on Pervasive Computing Technologies for Healthcare*. ACM, pp 108–117, <https://doi.org/10.1145/3240925.3240940>, URL <https://doi.org/10.1145/3240925.3240940>

Shi L, Liu F, Liu Y, et al (2023) Biofeedback respiratory rehabilitation training system based on virtual reality technology. *Sensors* 23(22):9025. <https://doi.org/10.3390/s23229025>, URL <https://doi.org/10.3390/s23229025>

Siebieszuk A, Płoński AF, Baranowski M (2025) Breathwork for chronic stress and mental health: Does choosing a specific technique matter? *Medical Sciences* 13(3):127. <https://doi.org/10.3390/medsci13030127>

Spee BT, Arato J, Mikuni J, et al (2024) The dynamics of experiencing gestalt and aha in cubist art: pupil responses and art evaluations show a complex interplay of task, stimuli content, and time course. *Frontiers in Psychology* 15:1192565. <https://doi.org/10.3389/fpsyg.2024.1192565>

Spehar B, van Tonder GJ (2017) Koffka’s aesthetic gestalt. *Leonardo* 50(1):53–57. https://doi.org/10.1162/leon_a.01020

Stepanova ER, Desnoyers-Stewart J, Pasquier P, et al (2020) Jel: Breathing together to connect with others and nature. In: *Proceedings of the 2020 ACM Designing Interactive Systems Conference*. ACM, pp 641–654, <https://doi.org/10.1145/3357236.3395532>, URL <https://doi.org/10.1145/3357236.3395532>

Stocker K, Hartmann M, Schmid Y, et al (2025) The 3d-ascr scale: A revalidation of the core dimensions of the altered states of consciousness rating scale 5d(11)-asc for psychedelic research. *Journal of Psychopharmacology* <https://doi.org/10.1177/02698811251397328>

Studerus E, Gamma A, Vollenweider FX (2010) Psychometric evaluation of the altered states of consciousness rating scale (oav). *PLOS ONE* 5(8):e12412. <https://doi.org/10.1371/journal.pone.0012412>

Suzuki K, Garfinkel SN, Critchley HD, et al (2013) Multisensory integration across exteroceptive and interoceptive domains modulates self-experience in the

2347 rubber-hand illusion. *Neuropsychologia* 51(13):2909–2917. [https://doi.org/10.1016/](https://doi.org/10.1016/j.neuropsychologia.2013.08.014)
2348 [j.neuropsychologia.2013.08.014](https://doi.org/10.1016/j.neuropsychologia.2013.08.014)
2349
2350 Taves A (2020) Mystical and other alterations in sense of self: An expanded frame-
2351 work for studying nonordinary experiences. *Perspectives on Psychological Science*
2352 15(3):669–690. <https://doi.org/10.1177/1745691619895047>, URL [https://doi.org/](https://doi.org/10.1177/1745691619895047)
2353 [10.1177/1745691619895047](https://doi.org/10.1177/1745691619895047)
2354
2355 Terhune DB, Luke DP, Kaelen M, et al (2016) A placebo-controlled investigation of
2356 synaesthesia-like experiences under LSD. *Neuropsychologia* 88:28–34. [https://doi.](https://doi.org/10.1016/j.neuropsychologia.2016.04.005)
2357 [org/10.1016/j.neuropsychologia.2016.04.005](https://doi.org/10.1016/j.neuropsychologia.2016.04.005)
2358
2359 Tinga AM, Nyklíček I, Jansen MP, et al (2019) Respiratory biofeedback does not facil-
2360 itate lowering arousal in meditation through virtual reality. *Applied Psychophysiol-*
2361 *ogy and Biofeedback* 44(1):51–59. <https://doi.org/10.1007/s10484-018-9421-5>, URL
2362 <https://doi.org/10.1007/s10484-018-9421-5>
2363
2364 Tsakiris M, Tajadura-Jiménez A, Costantini M (2011) Just a heartbeat away from
2365 one’s body: Interoceptive sensitivity predicts malleability of body-representations.
2366 *Proceedings of the Royal Society B: Biological Sciences* 278(1717):2470–2476. [https:](https://doi.org/10.1098/rspb.2010.2547)
2367 [//doi.org/10.1098/rspb.2010.2547](https://doi.org/10.1098/rspb.2010.2547)
2368
2369 Tsuchiya N, Wilke M, Frässle S, et al (2015) No-report paradigms: Extracting the
2370 true neural correlates of consciousness. *Trends in Cognitive Sciences* 19(12):757–770.
2371 <https://doi.org/10.1016/j.tics.2015.10.002>
2372
2373 Uthaug MV, Mason NL, Havenith MN, et al (2021) An experience with Holotropic
2374 Breathwork is associated with improvement in non-judgement and satisfaction with
2375 life while reducing symptoms of stress in a Czech-speaking population. *Journal of*
2376 *Psychedelic Studies* 5(3):176–189. <https://doi.org/10.1556/2054.2021.00193>, URL
2377 <https://doi.org/10.1556/2054.2021.00193>
2378
2379 Vicary S, Sperling M, von Zimmermann J, et al (2017) Joint action aesthetics. *PLOS*
2380 *ONE* 12(7):e0180101. <https://doi.org/10.1371/journal.pone.0180101>
2381
2382 Vidal J, Andreu CI, Wrzesien M, et al (2025) Inducing selflessness through
2383 a numadelic virtual reality experience: a preliminary study. *Virtual Reality*
2384 30(1). <https://doi.org/10.1007/s10055-025-01293-z>, URL [https://doi.org/10.1007/](https://doi.org/10.1007/s10055-025-01293-z)
2385 [s10055-025-01293-z](https://doi.org/10.1007/s10055-025-01293-z)
2386
2387 Wagemans J, Elder JH, Kubovy M, et al (2012) A century of Gestalt psychol-
2388 ogy in visual perception: I. perceptual grouping and figure–ground organization.
2389 *Psychological Bulletin* 138(6):1172–1217. <https://doi.org/10.1037/a0029333>, URL
2390 <https://doi.org/10.1037/a0029333>
2391
2392 Wald IY, Maimon A, Keniger de Andrade Gensas L, et al (2023) Breathing based
immersive interactions for enhanced agency and body awareness: A claustrophobia

motivated study. In: Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, pp 1–7, https://doi.org/10.1145/3544549.3585897 , URL https://doi.org/10.1145/3544549.3585897	2393 2394 2395 2396
Weng HY, Feldman JL, Leggio L, et al (2021) Interventions and manipulations of interoception. <i>Trends in Neurosciences</i> 44(1):52–62. https://doi.org/10.1016/j.tins.2020.09.010	2397 2398 2399 2400
Yoon JM, Im SC, Kim K (2022) Effects of diaphragmatic breathing and pursed lip breathing exercises on the pulmonary function and walking endurance in patients with chronic stroke: a randomised controlled trial. <i>International Journal of Therapy and Rehabilitation</i> 29(8):1–11. https://doi.org/10.12968/ijtr.2021.0027 , URL https://doi.org/10.12968/ijtr.2021.0027	2401 2402 2403 2404 2405 2406
Zaccaro A, Piarulli A, Laurino M, et al (2018) How breath-control can change your life: a systematic review on psycho-physiological correlates of slow breathing. <i>Frontiers in Human Neuroscience</i> 12:353. https://doi.org/10.3389/fnhum.2018.00353	2407 2408 2409 2410
Zaccaro A, Piarulli A, Melosini L, et al (2022) Neural correlates of non-ordinary states of consciousness in pranayama practitioners: the role of slow nasal breathing. <i>Frontiers in Systems Neuroscience</i> 16:803904. https://doi.org/10.3389/fnsys.2022.803904	2411 2412 2413 2414
Zhou Y, Sun B, Li Q, et al (2011) Sensitivity of bronchopulmonary receptors to cold and heat mediated by transient receptor potential cation channel subtypes in an ex vivo rat lung preparation. <i>Respiratory Physiology & Neurobiology</i> 177(3):327–332. https://doi.org/10.1016/j.resp.2011.05.011	2415 2416 2417 2418 2419 2420 2421 2422 2423 2424 2425 2426 2427 2428 2429 2430 2431 2432 2433 2434 2435 2436 2437 2438